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(54) **SCROLLING LASER ILLUMINATION WITH
A PHASE LIGHT MODULATOR**

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filed on Dec. 17, 2020.

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G02F 1/29 (2006.01)

(52) **U.S. Cl.**
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21/208 (2013.01)

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H04N 9/3126; H04N 9/3164
See application file for complete search history.

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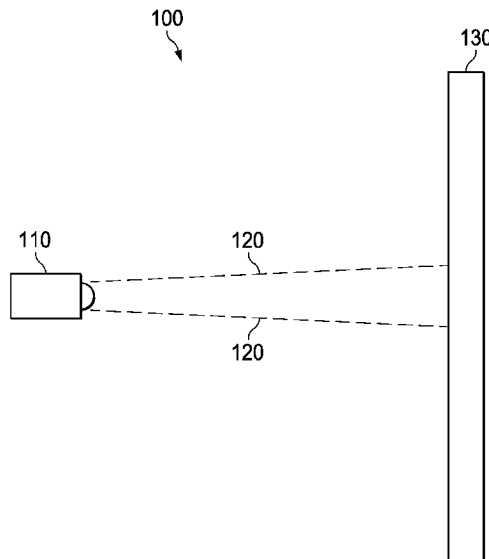
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(57) **ABSTRACT**
An apparatus includes a spatial light modulator (SLM)
having a first region and a second region and a phase light
modulator (PLM) optically coupled to the SLM. The PLM
is configured to receive first light having a first color and
receive second light having a second color, the second color
different than the first color. The PLM is also configured to
direct the first light towards the first region of the SLM and
direct the second light towards the second region of the
SLM.

20 Claims, 12 Drawing Sheets



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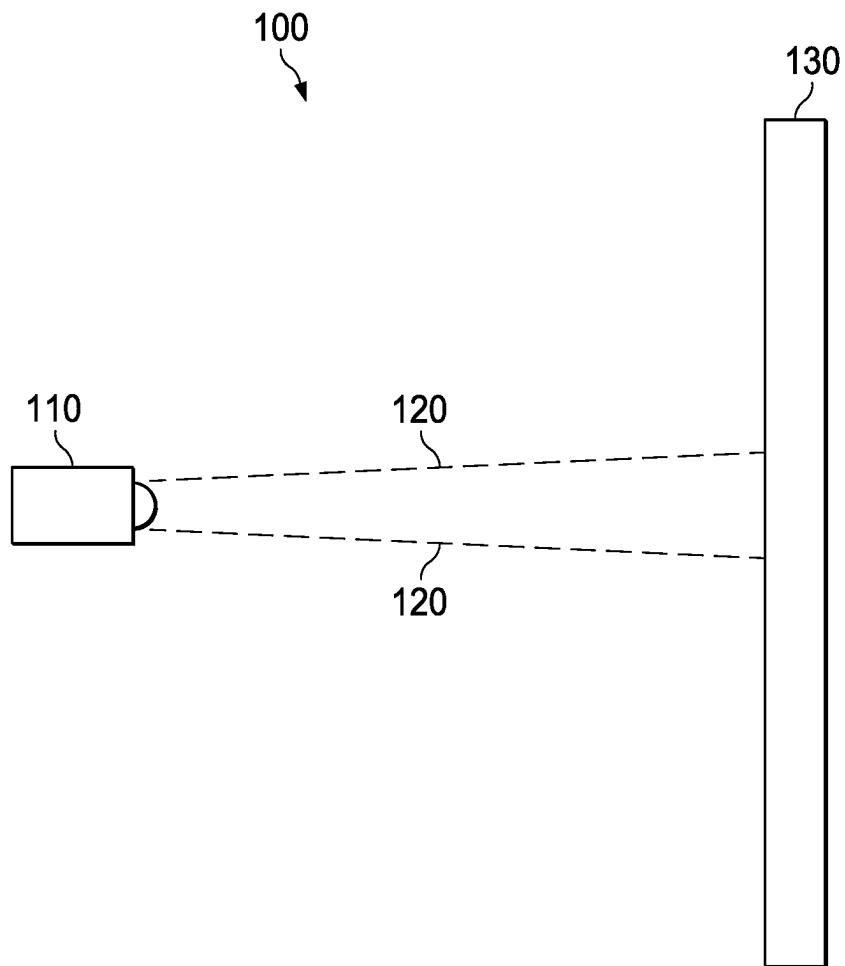


FIG. 1

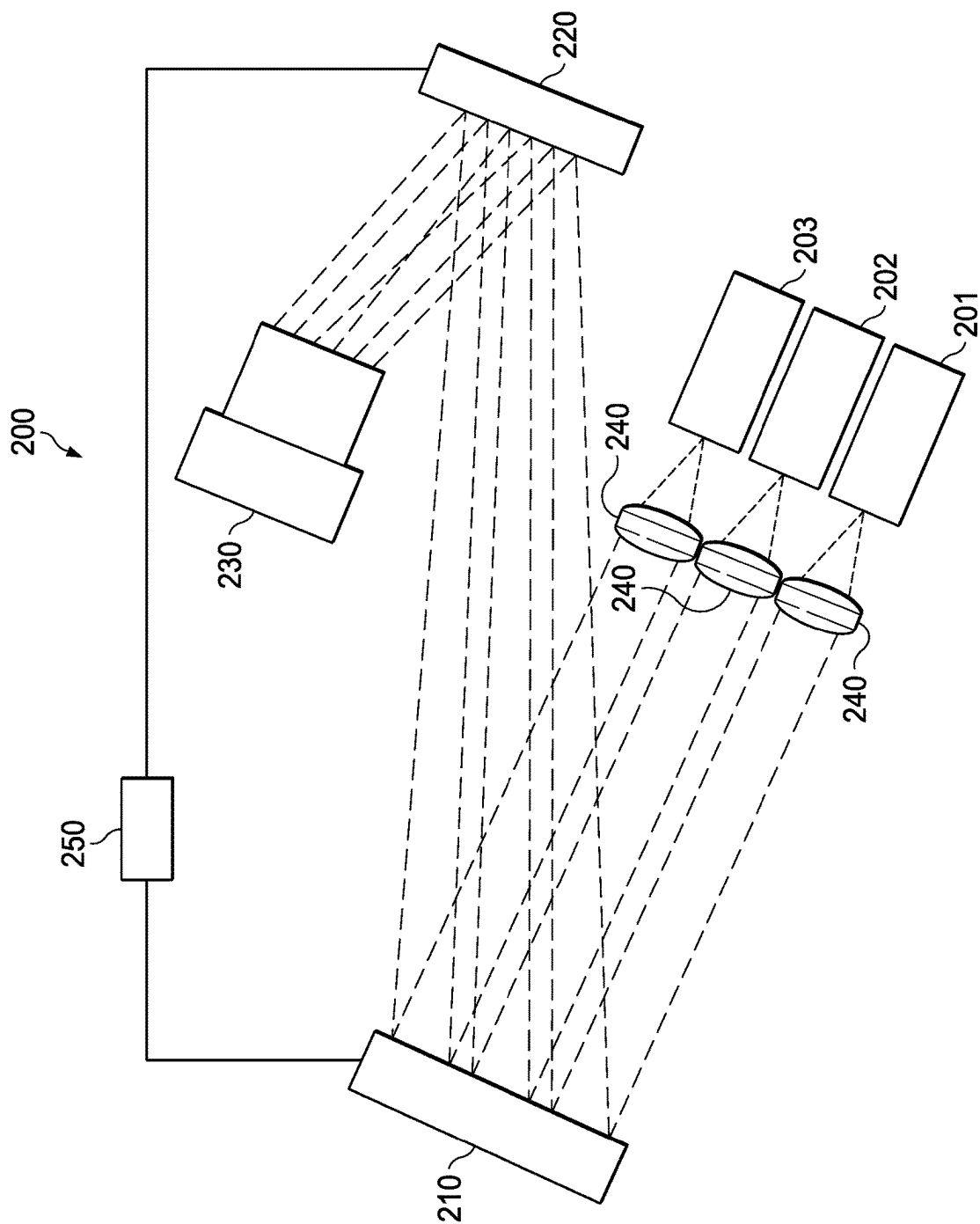
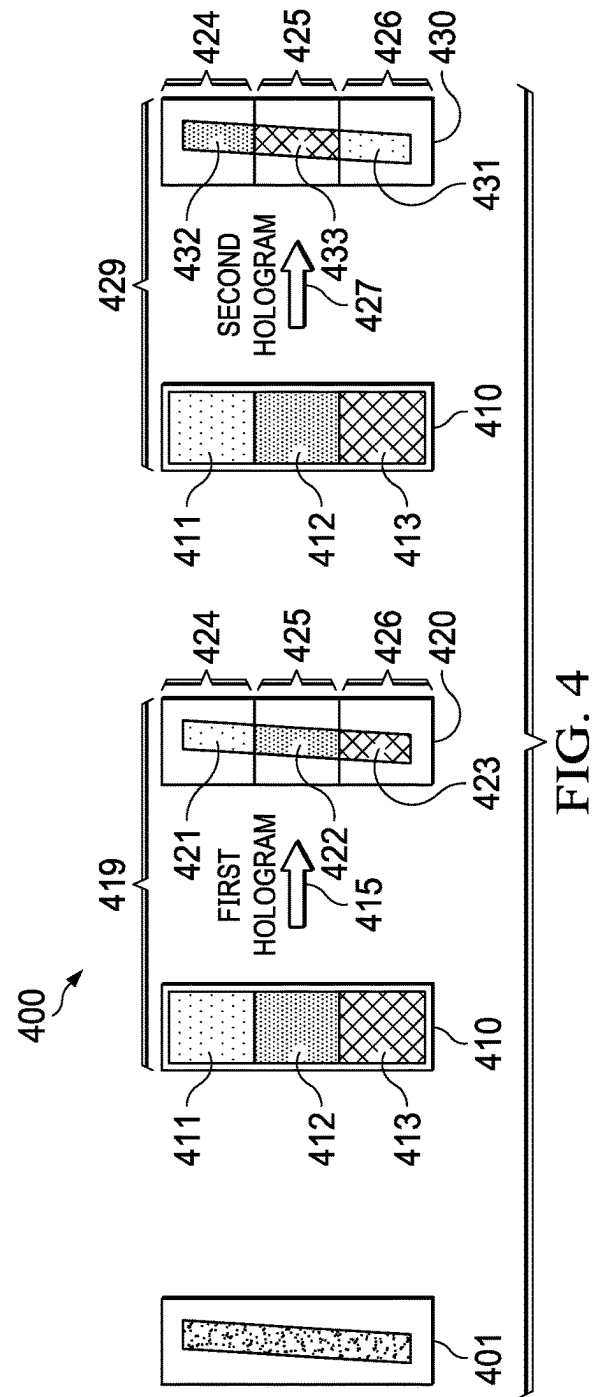
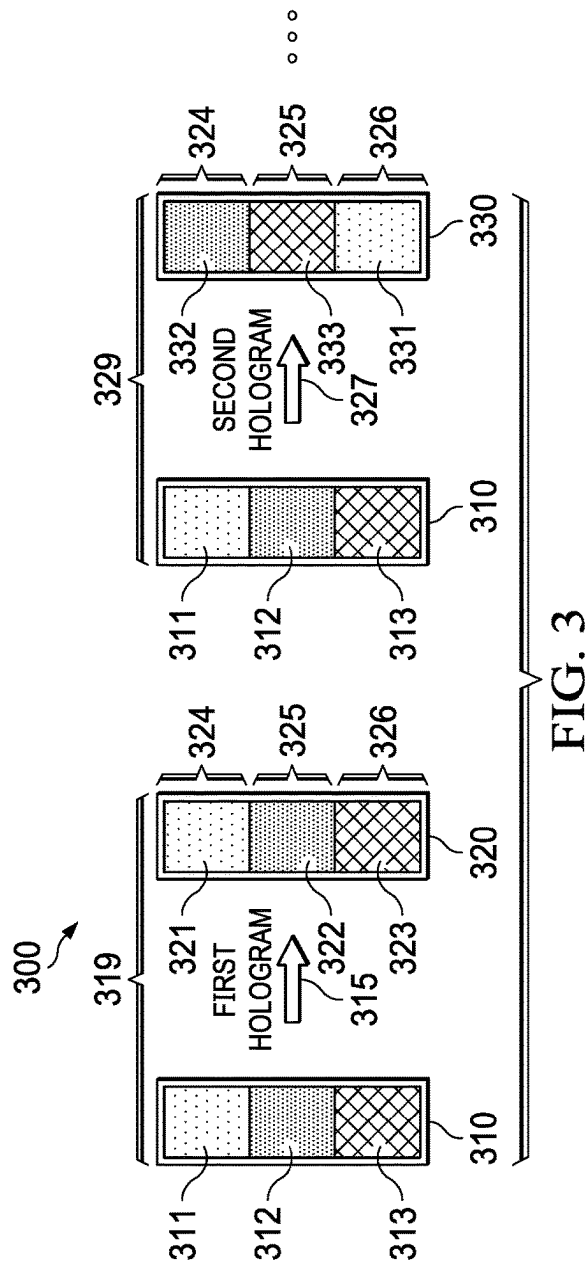


FIG. 2



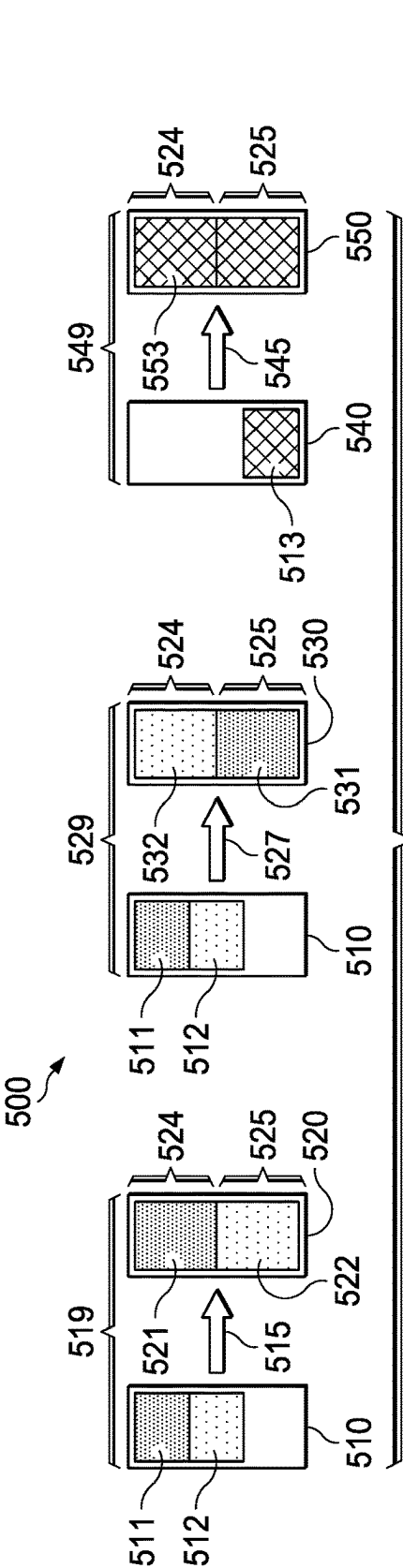


FIG. 5

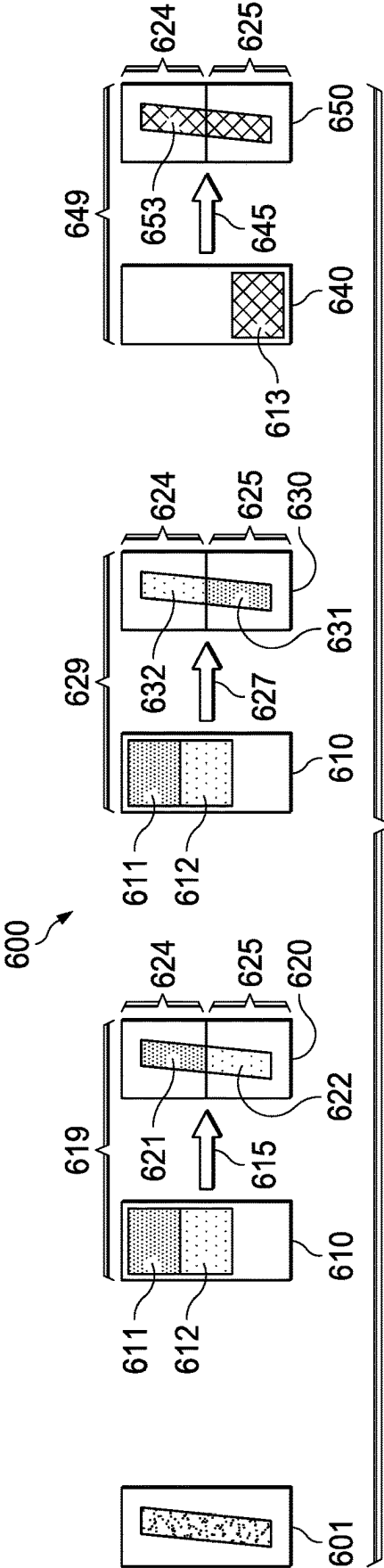


FIG. 6

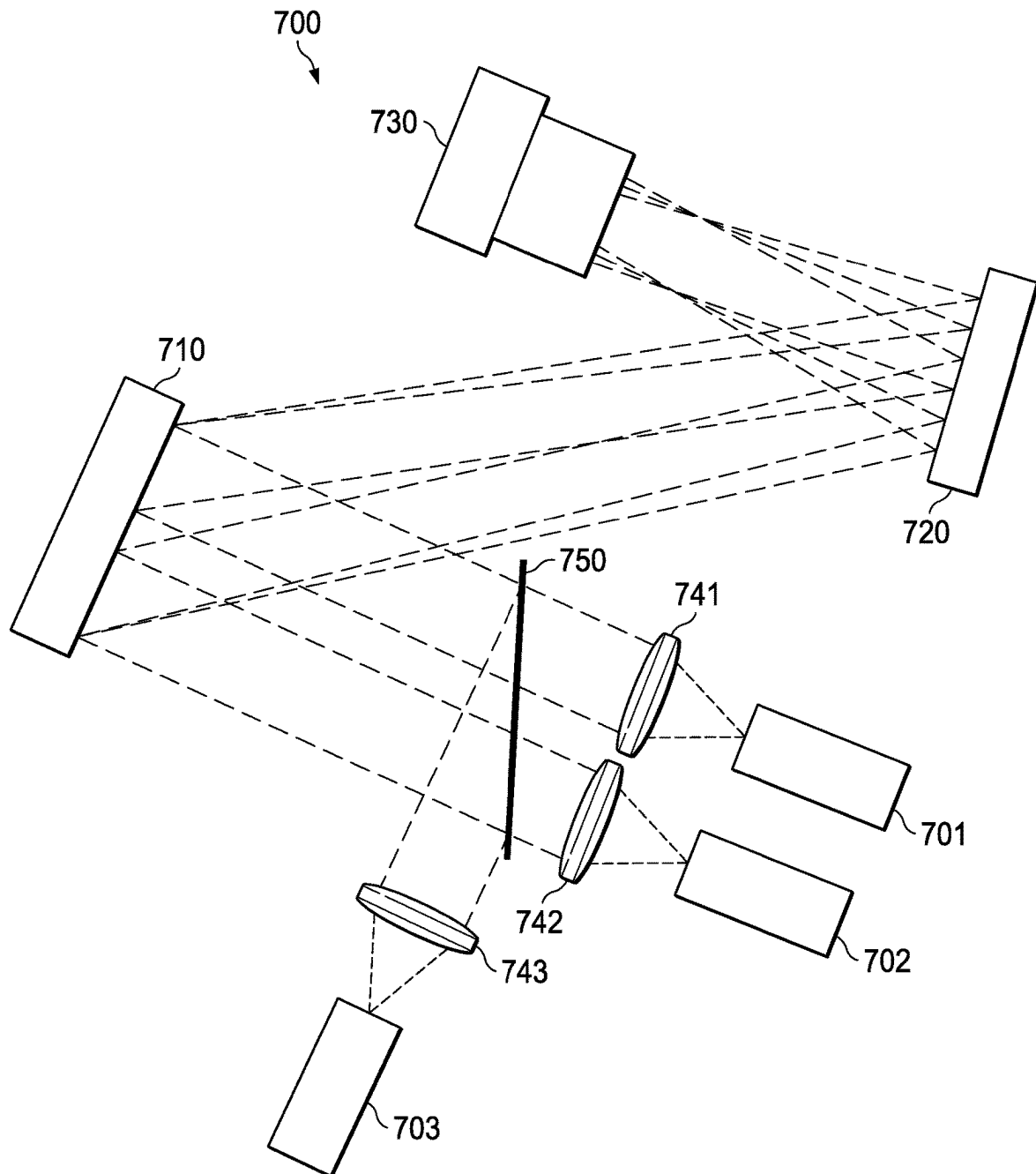
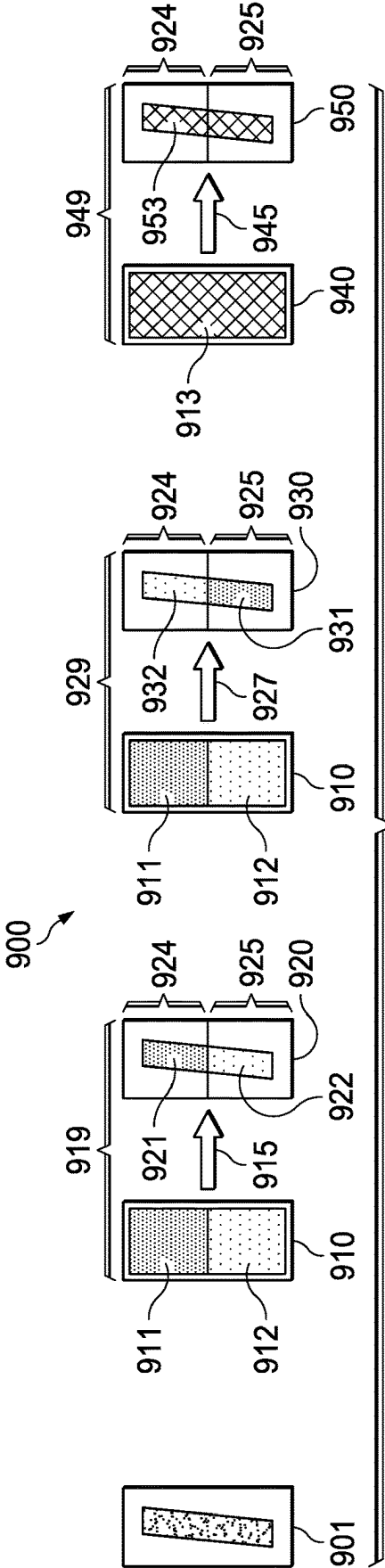
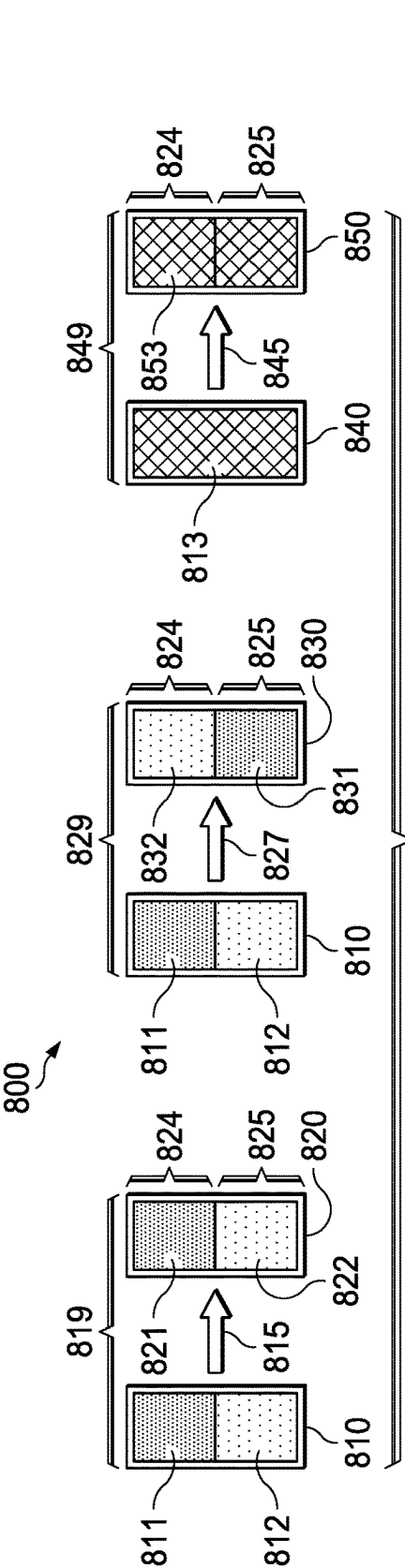


FIG. 7



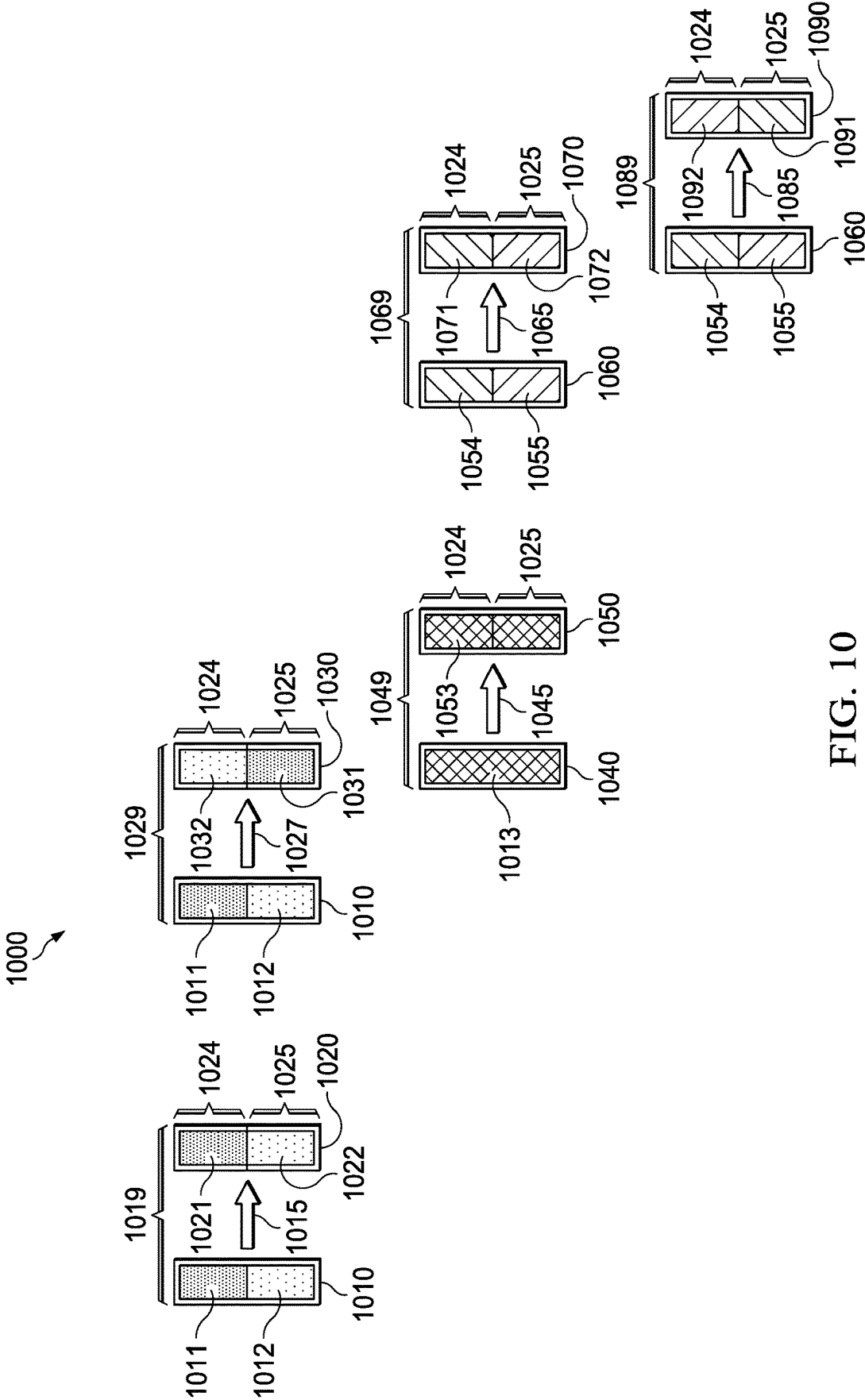
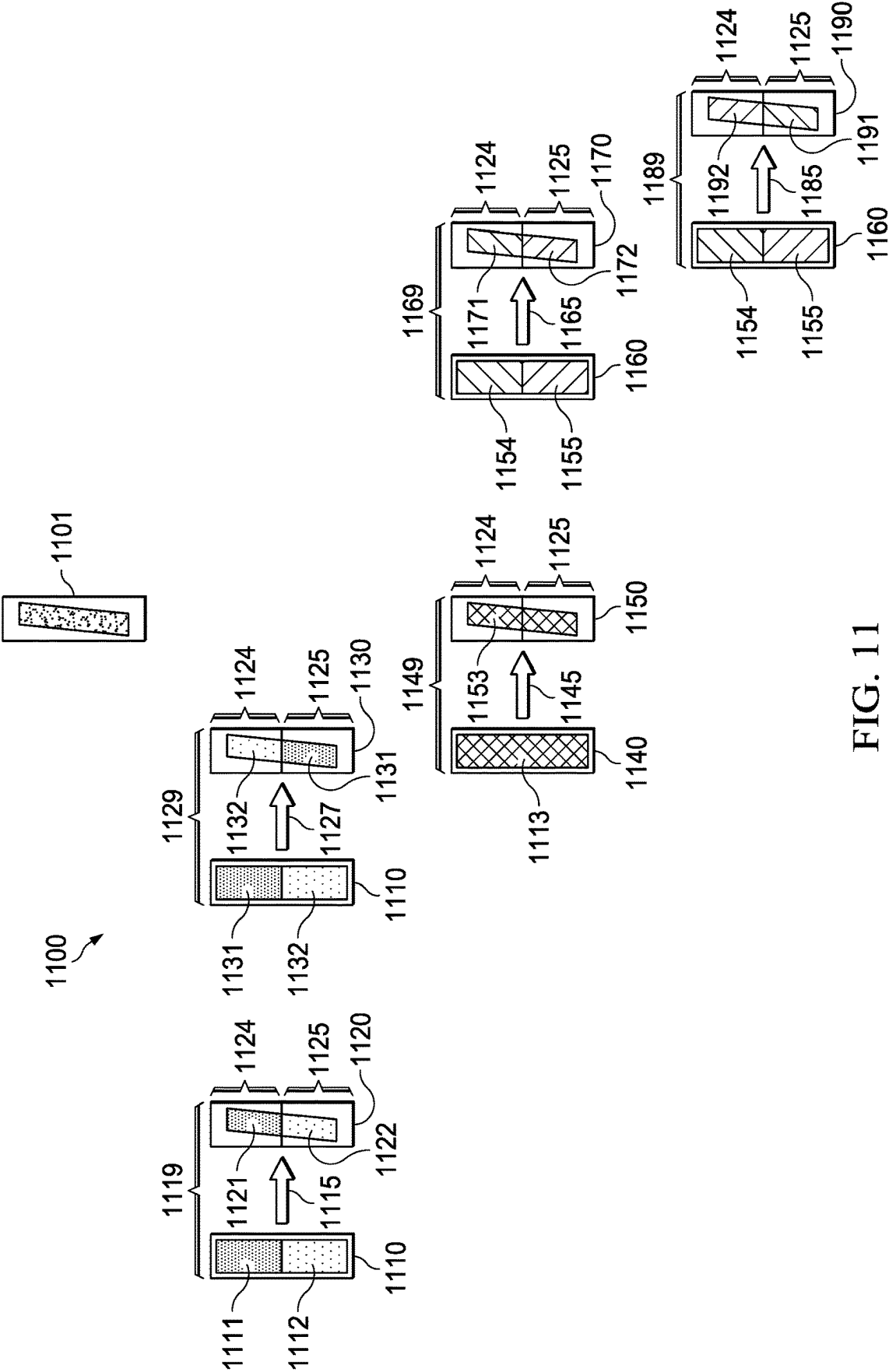
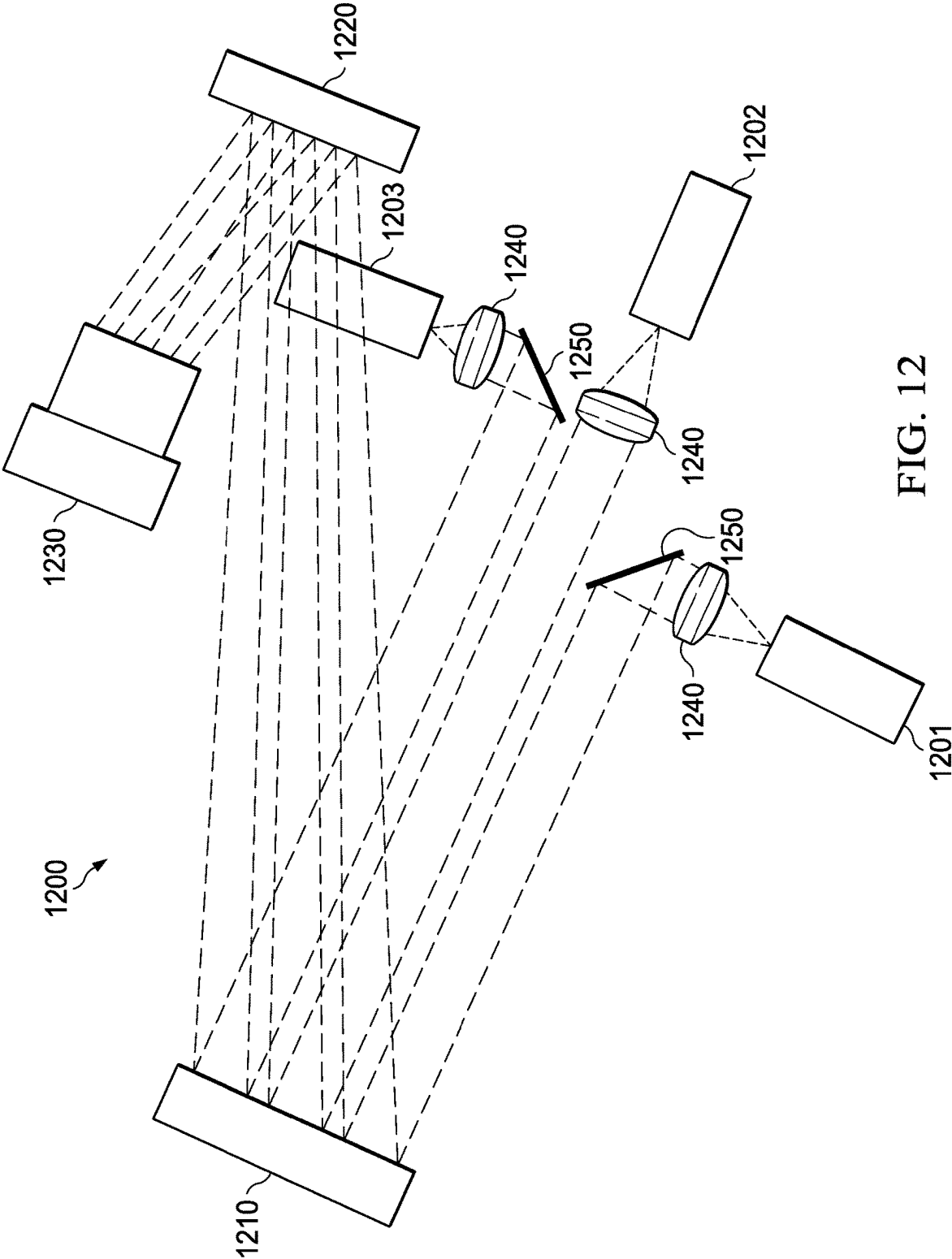


FIG. 10





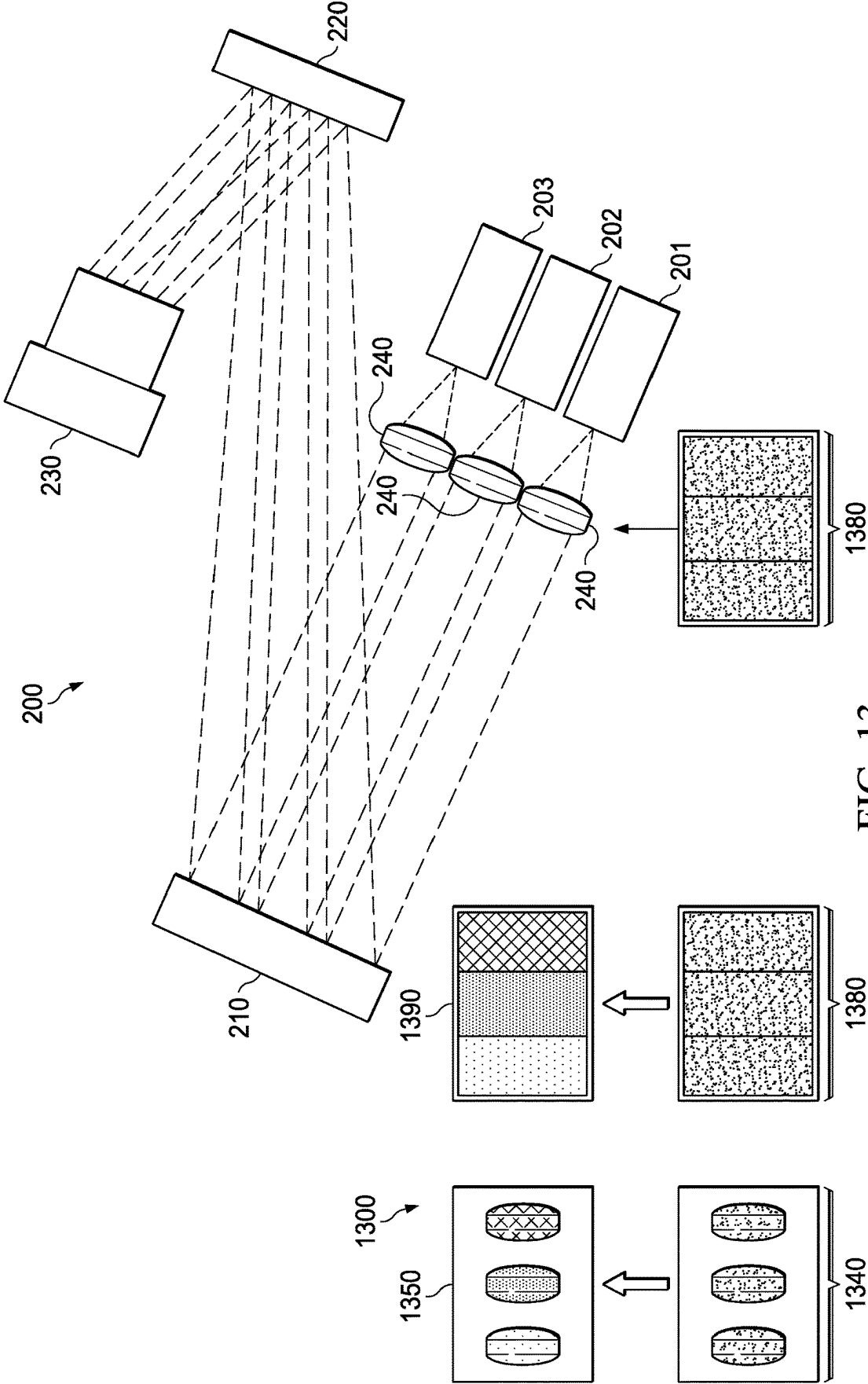


FIG. 13

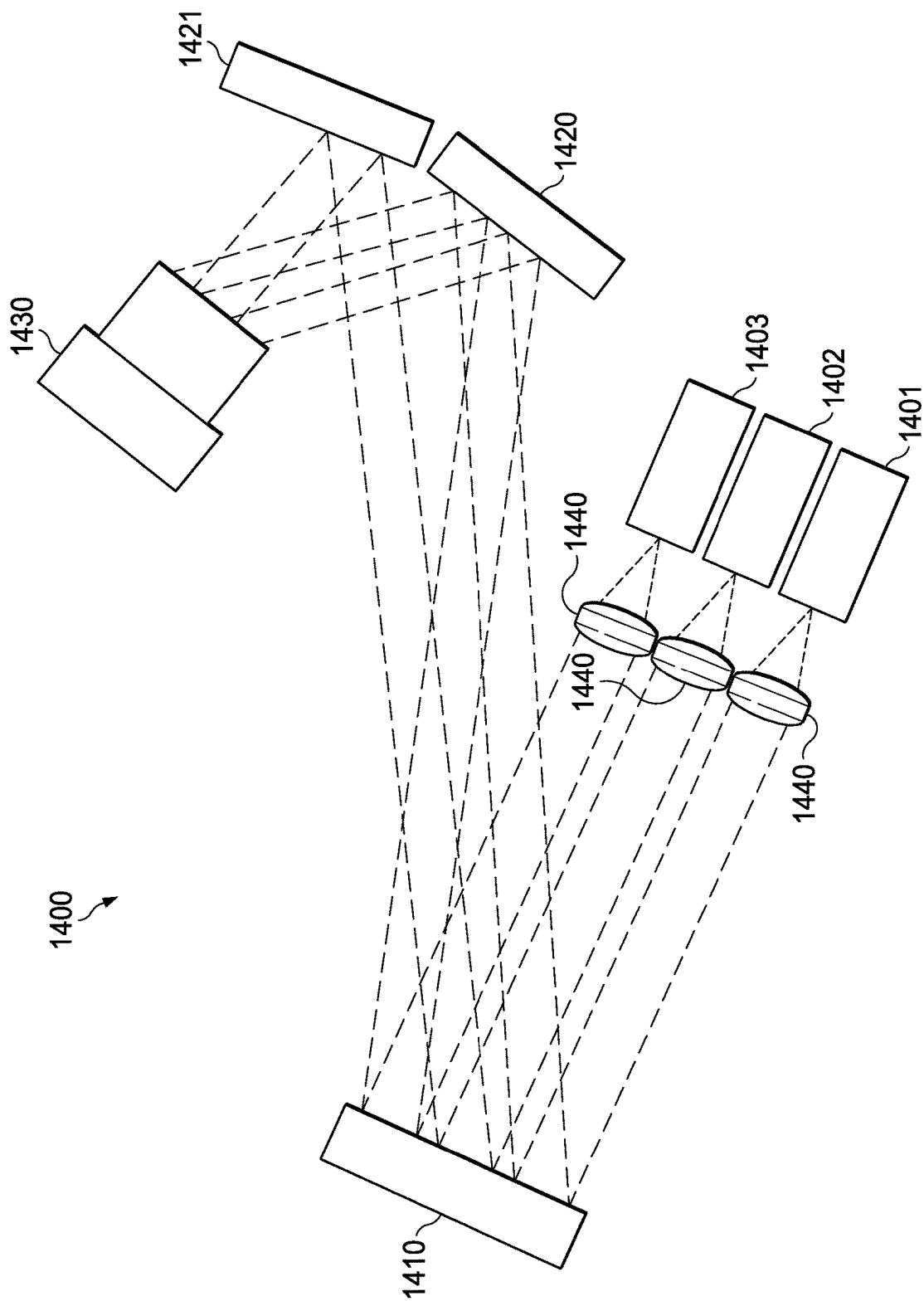


FIG. 14

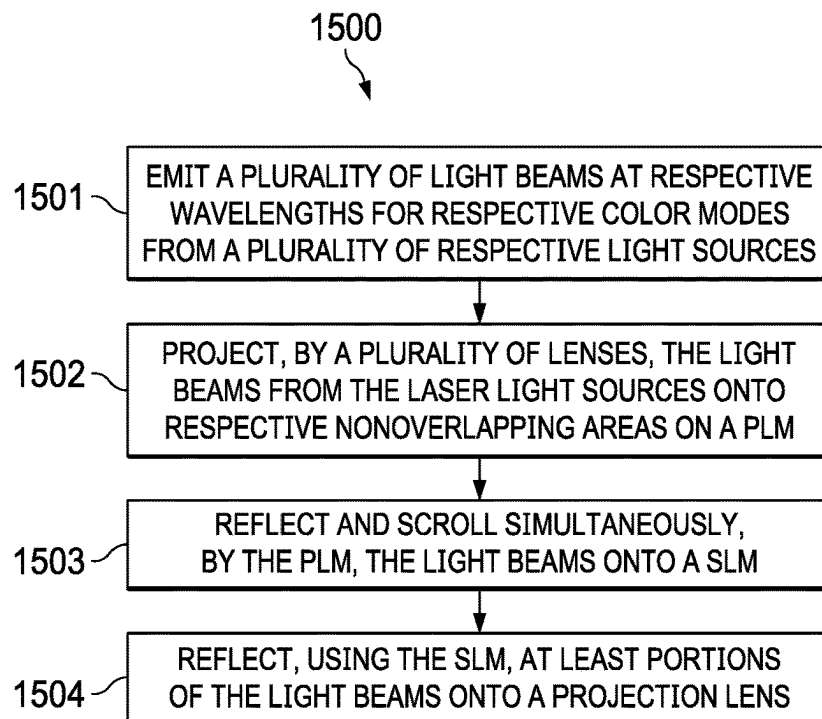


FIG. 15

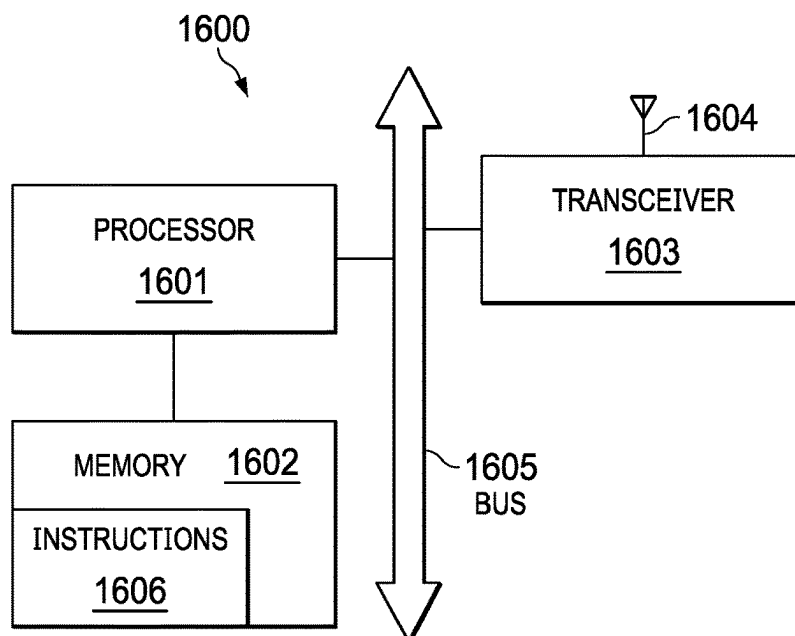


FIG. 16

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SCROLLING LASER ILLUMINATION WITH A PHASE LIGHT MODULATOR

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 17/539,039 filed Nov. 30, 2021, which claims priority to U.S. Provisional Patent Application No. 63/126,622, which was filed Dec. 17, 2020, entitled “Scrolling Laser Illumination Plus High Dynamic Range (HDR) Using A Phase Light Modulator (PLM),” and to U.S. Provisional Patent Application No. 63/273,076, which was filed Oct. 28, 2021, entitled, “Scrolling Laser Illumination with a Phase Light Modulator,” which Applications are hereby incorporated herein by reference in their entireties.

BACKGROUND

Projection-based displays project images onto surfaces such as onto a wall or a screen, to present video or still pictures. Such displays can include cathode-ray tube (CRT) displays, liquid crystal displays (LCDs), and digital mirror device (DMD) displays, etc. A display includes a display with adjustable display elements, which are usually arranged in a matrix of rows and columns. Each element of the display represents a pixel of an image that is to be projected. The elements are adjusted by an image processor to correspond to the brightness of the respective pixel in the image to be presented. The display system can also be based on one of various display schemes. In a time-sequential projection display system, light emitted by a light source, such as a lamp or light bulb, is separated by filters into basic color modes in a time sequence and projected on a single display. The sequence is set at a repetition rate sufficiently high to be combined by the human eye into a full image. Alternatively, in a scrolling color projection system, the color modes are projected and moved across the display at a sufficient scrolling rate to observe a full image by the human eye. The scrolling movement on the display is sometimes provided by a mechanical apparatus.

SUMMARY

In accordance with at least one example of the disclosure, an apparatus includes a spatial light modulator (SLM) having a first region and a second region and a phase light modulator (PLM) optically coupled to the SLM. The PLM is configured to receive first light having a first color and receive second light having a second color, the second color different than the first color. The PLM is also configured to direct the first light towards the first region of the SLM and direct the second light towards the second region of the SLM.

In accordance with another example of the disclosure, an apparatus includes a first spatial light modulator (SLM) and a second SLM. The apparatus also includes a phase light modulator (PLM) optically coupled to the first SLM and to the second SLM. The PLM is configured to receive first light having a first color and receive second light having a second color, the second color different than the first color. The PLM is also configured to direct the first light towards the first SLM and direct the second light towards the second SLM.

In accordance with another example of the disclosure, a method includes receiving, by a first region of a phase light modulator (PLM), first light having a first color and receiv-

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ing, by a second region of the PLM, second light having a second color, the second color different from the first color. The method also includes producing, by the PLM, a holographic image responsive to receiving the first light and the second color, where the holographic image has a first region and a second region, the first region of the holographic image having the first color and the second region of the holographic image having the second color.

BRIEF DESCRIPTION OF THE DRAWINGS

For a detailed description of various examples, reference will now be made to the accompanying drawings in which:

FIG. 1 is a block diagram of a display system, in accordance with various examples.

FIG. 2 is a block diagram of a display device, in accordance with various examples.

FIG. 3 is a block diagram of illumination projections in the display device of FIG. 2, in accordance with various examples.

FIG. 4 is a block diagram of illumination projections of an HDR image pattern in the display device of FIG. 2, in accordance with various examples.

FIG. 5 is a block diagram of illumination projections in the display device of FIG. 2, in accordance with various examples.

FIG. 6 is a block diagram of illumination projections of an HDR image pattern in the display device of FIG. 2, in accordance with various examples.

FIG. 7 is a block diagram of a display device, in accordance with various examples.

FIG. 8 is a block diagram of illumination projections in the display device of FIG. 7, in accordance with various examples.

FIG. 9 is a block diagram of illumination projections of an HDR image pattern in the display device of FIG. 7, in accordance with various examples.

FIG. 10 is a block diagram of illumination projections in the display device of FIG. 7, in accordance with various examples.

FIG. 11 is a block diagram of illumination projections of an HDR image pattern in the display device of FIG. 7, in accordance with various examples.

FIG. 12 is a block diagram of a display device, in accordance with various examples.

FIG. 13 is a block diagram of illumination patterns of a display device, in accordance with various examples.

FIG. 14 is a block diagram of a display device, in accordance with various examples.

FIG. 15 is a flow diagram of a method for light projection using a PLM, in accordance with various examples.

FIG. 16 is a block diagram of a processing system, in accordance with various examples.

DETAILED DESCRIPTION

A projection-based display system may include an SLM device which includes optical elements, such as mirrors or apertures, to generate an image. An SLM modulates the intensity of the light beam projected on the display by controlling the optical elements to manipulate the light and form the pixels of an image. The SLM may be a DMD in which the optical elements are tilting micromirrors. Each micromirror projects a pixel of the image to be displayed. The micromirrors are tilted by applying voltages to the micromirrors to project dark, bright, or shades of light per pixel.

A projection-based display system may also include multiple light sources, such as laser light sources, of different wavelengths to provide the color modes rather than a single lamp or light bulb. The light sources can be operated by simultaneously projecting the color modes and scrolling light beams from the light sources across the SLM surface in time to form the image. The scrolling mechanism is sometimes achieved via mechanical mechanisms to scroll the light beams across the SLM. However, mechanical mechanisms for scrolling the light beams can introduce design challenges such as large size or high noise.

This description describes various examples of a PLM device to scroll light beams from respective light sources for a projection-based display system. A PLM may be a micro-electromechanical system (MEMS) device including micromirrors that have adjustable heights with respect to the PLM surface. The heights of the micromirrors can be adjusted by applying voltages. The micromirrors may be controlled, such as in groups, with different voltages to form a diffraction surface on the PLM which is a phase altering reflective surface to the incident light beams from the light sources. The phase altering reflective surface may form holograms for projecting illumination patterns that form an image on a projection surface. The holograms are formed by adjusting the micromirrors to form the diffraction surface of the PLM. The micromirrors of the PLM may be further controlled by changing the voltages applied to the micromirrors to modify the diffraction surface and accordingly the hologram which changes the angles of reflected light beams with respect to the surface of the PLM. Changing the angles of the light beams causes the scrolling of the light beams on the projection surface.

In comparison to other devices with mechanical based components for light scrolling, the PLM provides the scrolling of the light beams without a mechanical apparatus and accordingly allows for reducing the device size and cost and the signal noise level. A PLM device may also allow the overlapping of light beams of different color modes providing additional color modes, as described in examples below. Accordingly, few laser light sources may be included in the device to provide a greater number of color modes. For example, the device may include two or three laser light sources and provide more than three color modes for image display. Overlapping the color modes may also increase color gamut flexibility. The PLM may also be operated according to HDR modulation techniques, as described in examples below, to increase brightness in images. The illumination areas of different color modes may also be increased, as described in examples below, which may be useful to compensate for the laser light source ability to provide sufficient light, also referred to herein as Etendue of the laser light source. Increasing the illumination area and accordingly the Etendue of the laser light source may increase the resolution of the image being projected by the device. The light sources may be turned on continuously to illuminate different portions of the PLM at different color modes resulting in higher average output power and thus brighter images.

According to various examples, the projection surface may be an SLM or a display. The light beams corresponding to the color modes can be scrolled by the PLM onto an SLM, which in turn projects light onto a display. A color mode of the light beam may depend on the light source of the light beam. The color mode may be the visible color of the light emitted by the light source, as perceived by a human eye, in accordance with the wavelength, a wavelength range, or a combination of wavelengths of the emitted light. Alternately,

the light beams may be scrolled by the PLM directly onto a display without an SLM in the projection-based display system. The projection-based display system may also include a projection lens between the SLM or PLM and the display, and lenses between the light sources and the PLM. The lenses adjust the illumination patterns of the light beams onto the surface of the PLM. In other examples, the system may include multiple PLMs for reflecting the respective color modes from the light sources. Each PLM may reflect a light beam of a color mode from a respective light source onto the SLM, the projection lens or the display. The system may also use HDR modulation techniques to spatially modulate the light distribution on the PLM or SLM to cause brighter areas in the image to receive more light intensity.

In examples, the system may include multiple SLMs, such as two or three SLMs, for projecting the light beams of different color modes provided by respective light sources. For example, the PLM may scroll one or two light beams for two respective color modes onto a first SLM, and a third light beam for a third color mode onto a second SLM. The first and second SLMs may project the light beams for the three color modes onto the projection surface. In another example, the PLM may scroll the light beam for each color mode onto a respective SLM. The color modes may be for blue, green, and red lights. Accordingly, the PLM may scroll a first light beam for the blue light onto a first SLM, a second light beam for the green light onto a second SLM, and a third light beam for the red light onto a third SLM. The three SLMs may project the respective light beams for the blue, green, and red lights onto the projection surface to form a full color image.

FIG. 1 is a block diagram of a display system 100, in accordance with various examples. The display system 100 may be a projection-based display system for projecting images or video. The display system 100 includes a display projector 110 configured to project light 120 onto a projection display 130, such as a screen or a wall. The light 120 may be controlled or modulated by the display projector 110 to project still images or moving images, such as video, on the projection display 130. The light 120 may be formed as a combination of light beams corresponding to multiple color modes provided by the display projector 110. The display projector 110 may include light sources (not shown) for providing the light beams at different wavelengths. The display projector 110 may also include optical components (not shown) for controlling the light beams to providing the images or video on the projection display 130. The display projector 110 may also include a controller (not shown) for controlling the components of the display projector 110 to display the images or video.

FIG. 2 is a block diagram of a display device 200 which may be part of the display projector 110, in accordance with various examples. The display device 200 may include light sources 201, 202, and 203, a PLM 210, an SLM 220, and a projection lens 230. The light sources 201, 202, and 203 may be any light emitting devices that emit respective light beams directed toward the PLM 210. The light sources 201, 202, and 203 may emit light in different wavelengths of different color modes. For example, the light sources 201, 202, and 203 may be laser light sources that emit light in the visible spectrum. The different light wavelength of the light sources 201, 202, and 203 may correspond to different colors of visible light, such as red, green, and blue. The display device 200 may also include lenses 240 positioned between the light sources 201, 202, and 203 and the PLM 210. Each one of the light sources 201, 202, and 203 may be aligned

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with a respective lens **240**. The display device **200** may also include a controller **250** configured to control the operations of the PLM **210** and the SLM **220**. The controller **250** may also control switching on and off the light sources **201**, **202**, and **203**.

The PLM **210** may be configured to reflect the projected light beams from the light sources **201**, **202**, and **203** onto the SLM **220**. The PLM **210** may be controlled by a controller **250** to control the reflected light beams to the SLM **220** and provide illumination patterns on the surface of the SLM **220**. The illumination patterns provide an image projected by the projection lens **230**. The PLM **210** may include an array of micromirrors positioned on the surface of the PLM **210** and having adjustable heights with respect to the surface responsive to the voltage. The PLM **210** may be controlled by the controller **250** to provide variation in the heights of the micromirrors across the surface of the PLM **210**. The heights of the micromirrors may be changed in steps, according to the applied voltage levels, to provide a phase variation on the surface of the PLM **210**. The steps in heights may be provided according to the electronic components of the display projector **110**. For example, the electronic components may include digital circuits with a bit size that determines the increment in steps or the number of steps. The variation in the heights of the micromirrors forms a diffraction surface on the PLM **210**. The diffraction surface may be programmed to provide a hologram at the surface of the PLM **210** for reflecting the light beams from the light sources **201**, **202**, and **203** onto the SLM **220**. The reflected light beams form, on the surface of the SLM **220**, illumination patterns on the pixels of the SLM **220**. The image may be formed by controlling the illuminated micromirrors of the SLM **220**. The voltages applied to the PLM **210** may be varied in time, changing the diffraction surface of the PLM **210** to scroll the reflected light beams and scan the pixels on the SLM **220** that form the image. The micromirrors of the PLM **210** may be controlled to scroll the light beams onto the SLM **220** and illuminate the pixels of the SLM **220**.

The SLM **220** may reflect the light beams from the PLM **210** onto the projection lens **230** to project the image. The SLM **220** may be controlled by the controller **250** to control the reflected light beams to the projection lens **230**, which in turn projects the image to the projection display **130**. The SLM **220** may include an array of tilting micromirrors positioned on the surface of the SLM **220**. The micromirrors may be arranged in a 2D array on the surface of the SLM **220** and have two states (e.g., on and off) that can be switched by applying two respective voltage levels. Each micromirror of the SLM **220** may project a pixel of the image onto the projection lens **230**. In the on state, the micromirrors of the SLM **220** may be tilted to a first angle to reflect the light beams from the PLM **210** to the projection lens **230**. In the off state, the micromirrors may be tilted to a second angle to direct the light beams away from the projection lens **230**, such as onto a heatsink or a light dump, making the respective pixels at the projection lens **230** appear dark. The micromirrors may be toggled on and off at a rate to cause the human eye to perceive shades of light at the pixels. The toggle ratio of on to off states may determine the shade of light provided at the pixel.

The projection lens **230** may include one or more lenses (not shown) to properly align the light beams from the SLM **220** onto the projection display **130** and project the light **120** on the projection display **130**. The projection lens **230** may

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also include other optical components (not shown) for projecting aligning the light beams and projecting the light **120**.

The lenses **240** are designed for projecting the respective light beams from the light sources **201**, **202**, and **203** onto a surface of the PLM **210**. The lenses **240** may be designed to provide illumination patterns of the light beams on the surface of the PLM **210** in accordance with the illumination requirement for the images to be displayed. The lenses **240** may be positioned at respective distances and locations from the light sources **201**, **202**, and **203** to provide nonoverlapping illumination patterns of the light beams on the surface of the PLM **210**. The light beams may have light intensity profiles based on the respective light sources **201**, **202**, and **203**, such as a Gaussian profile. In examples, the lenses **240** may be configured and positioned to provide uniform illumination patterns of the light beams on the surface of the PLM **210**. Responsive to applying voltages to the PLM **210** to form a diffraction surface and a respective hologram for reflecting the light beams, the illumination patterns on the surface of the PLM **210** are projected on the surface of the SLM **220** providing respective illumination patterns of the image.

FIG. 3 is a block diagram of illumination projections **300** in the display device **200**, in accordance with various examples. The illumination projections **300** may be projected in a sequence of image subframes at a time rate causing the human eye to combine the image subframes into a single image. For example, the illumination projections **300** may be projected in a sequence of three or more subframes. The illumination projections **300** include a first illumination projection **310** of the light beams from the light sources **201**, **202**, and **203** on the PLM **210**. The first illumination projection **310** includes first illumination patterns **311**, **312**, and **313** of the light sources **201**, **202**, and **203**, respectively. The first illumination patterns **311**, **312**, and **313** may have nonoverlapping illumination areas on the PLM **210**. The first illumination patterns **311**, **312**, and **313** may have a first arrangement on the surface of the PLM **210** where the color mode of the light source **202** is positioned between the respective color modes of the light sources **201** and **203**, respectively. The PLM **210** may be controlled according to a first voltage setting to form a first hologram **315**, which reflects the light beams from the PLM **210** providing, in a first subframe **319**, a second illumination projection **320** of the light beams on the SLM **220**. The second illumination projection **320** includes second illumination patterns **321**, **322**, and **323** of the first hologram **315** produced by the PLM **210** based on the light beams from the respective light sources **201**, **202**, and **203**. The second illumination patterns **321**, **322**, and **323** may have nonoverlapping illumination areas on the SLM **220**. The second illumination patterns **321**, **322**, and **323** may be projected on respective pixels or pixel portions **324**, **325**, and **326** at the SLM **220**. According to the first hologram **315**, the second illumination patterns **321**, **322**, and **323** may be projected to match the first arrangement of the first illumination patterns **311**, **312**, and **313** on the PLM **210**, where the color mode of the light source **202** is projected at the pixel or pixel portion **325** between the respective color modes of the light sources **201** and **203** at the pixels or pixel portions **324** and **326**, respectively.

To scroll the light beams on the SLM **220**, the PLM **210** may be controlled according to a second voltage setting to form a second hologram **327**, which provides in a second subframe **329** a third illumination projection **330** of the light beams on the SLM **220**. The second hologram **327** causes

the light beams of the first illumination patterns **311**, **312**, and **313** to be reflected from the PLM **210** to the SLM **220** at different angles than the first hologram **315** thereby scrolling the light beams on the surface of the SLM **220**. The third illumination projection **330** includes third illumination patterns **331**, **332**, and **333** of the light beams of the respective light sources **201**, **202**, and **203**. The third illumination patterns **331**, **332**, and **333** may have uniform and nonoverlapping illumination areas on the SLM **220**. According to the second hologram **327**, the third illumination patterns **331**, **332**, and **333** may be projected in a second arrangement different than the first arrangement of the first illumination patterns **311**, **312**, and **313** on the surface of the PLM **210**. The third illumination patterns **331**, **332**, and **333** may be projected on the SLM **220** where the color mode of the light source **203** is projected at the pixel or pixel portion **325** between the respective color modes of the light sources **202** and **201** at the pixels or pixel portions **324** and **326**, respectively. Switching from the first hologram **315** to the second hologram **327** provides the scrolling of each of the light beams of the light sources **201**, **202**, and **203** from the second illumination projection **320** to the third illumination projection **330**. The scrolling of the light beams on the SLM **220** may be at a rate allowing the human eye to superimpose the second illumination patterns **321**, **322**, and **323** and the third illumination patterns **331**, **332**, and **333** with additional illumination patterns provided in subsequent subframes in the same pixels or pixel portions **324**, **325**, and **326** to perceive a single image.

FIG. **4** is a block diagram of illumination projections **400** of an HDR image pattern **401** in the display device **200**, in accordance with various examples. The illumination projections **400** may provide HDR images having brighter illumination and higher contrast, such as between bright and dark image portion within a pixel, in comparison to the images provided by the illumination projections **300** without HDR modulation. According to the HDR modulation, the scrolling of the light beams by the PLM **210** may be restricted to surface areas of the SLM **220** where the image is bright, excluding other areas. The HDR images may be provided by HDR modulation techniques for controlling the PLM **210**, the SLM **220**, or both. For example, the HDR image pattern **401** may be provided by an HDR modulation technique performed while controlling the light beams by the PLM **210** or both the PLM and SLM **220**. According to HDR modulation, the light beams may be reflected and scrolled by the PLM **210** onto the SLM **220** to illuminate certain regions of the pixels and away from other regions. Restricting the illumination to regions of the pixels as such may display an image with higher contrast, where the illuminated pixel regions may project bright areas of the image while the remaining areas may appear dark.

The illumination projections **400** includes a first illumination projection **410** of the light beams from the light sources **201**, **202**, and **203** on the PLM **210**. The first illumination projection **410** includes first illumination patterns **411**, **412**, and **413** of the light sources **201**, **202**, and **203**, respectively. The first illumination patterns **411**, **412**, and **413** may have uniform and nonoverlapping illumination areas on the PLM **210**. The first illumination patterns **411**, **412**, and **413** may be projected in a first arrangement on the PLM **210** where the color mode of the light source **202** is positioned between the respective color modes of the light sources **201** and **203**, respectively. The PLM **210** may be controlled according to a first voltage setting to form a first hologram **415**, which reflects the light beams from the PLM **210** providing, in a first subframe **419**, a second illumination

projection **420** of the light beams on the SLM **220**. The second illumination projection **420** includes second illumination patterns **421**, **422**, and **423** of the light beams from the respective light sources **201**, **202**, and **203**. The second illumination patterns **421**, **422**, and **423** may have uniform and nonoverlapping illumination areas on the SLM **220**. The second illumination patterns **421**, **422**, and **423** may be projected on respective pixels or pixel portions **424**, **425**, and **426** at the SLM **220**. The second illumination patterns **421**, **422**, and **423** may include HDR level contrast in each pixel portion **424**, **425**, and **426** where the image portions may be brighter than and illuminated without non-image portions in each pixel according to the HDR image pattern **401**. According to the first hologram **415**, the second illumination patterns **421**, **422**, and **423** may be projected to match the color distribution of the first illumination patterns **411**, **412**, and **413** on the PLM **210**, where the color mode of the light source **202** is projected at the pixel or pixel portion **425** between the respective color modes of the light sources **201** and **203** at the pixels or pixel portions **424** and **426**, respectively. The spatial distribution of the second illumination patterns **421**, **422**, and **423** may match the HDR image pattern **401**.

To scroll the light beams on the SLM **220**, the PLM **210** may be controlled according to a second voltage setting to form a second hologram **427**, which provides in a second subframe **429** a third illumination projection **430** of the light beams on the SLM **220**. The second hologram **427** causes the light beams of the first illumination patterns **411**, **412**, and **413** to be reflected from the PLM **210** to the SLM **220** at different angles than the first hologram **415** thereby scrolling the light beams on the surface of the SLM **220**. The third illumination projection **430** includes third illumination patterns **431**, **432**, and **433** of the light beams formed by the second hologram **427** produced by PLM **210** from the respective light sources **201**, **202**, and **203**. The third illumination patterns **431**, **432**, and **433** may have uniform and nonoverlapping illumination areas on the SLM **220**. According to the second hologram **427**, the third illumination patterns **431**, **432**, and **433** may be projected in a second arrangement different than the first arrangement of the first illumination patterns **411**, **412**, and **413** on the PLM **210**. The third illumination patterns **431**, **432**, and **433** may be projected on the SLM **220** where the color mode of the light source **203** is projected at the pixel or pixel portion **425** between the respective color modes of the light sources **202** and **201** at the pixels or pixel portions **424** and **426**, respectively. The third illumination patterns **431**, **432**, and **433** may include HDR level contrast in each pixel portion **424**, **425**, and **426** where the image portions may be brighter than and illuminated without non-image portions in each pixel according to the HDR image pattern **401**. Switching from the first hologram **415** to the second hologram **427** provide the scrolling of each of the light beams of the light sources **201**, **202**, and **203** from the second illumination projection **420** to the third illumination projection **430**. The scrolling of the light beams on the SLM **220** may be at a rate allowing the human eye to superimpose the second illumination patterns **421**, **422**, and **423** and the third illumination patterns **431**, **432**, and **433** with additional illumination patterns provided in subsequent subframes in the same pixels or pixel portions **424**, **425**, and **426** to perceive a single image.

FIG. **5** is a block diagram of illumination projections **500** in the display device **200**, in accordance with various examples. The illumination projections **500** may be projected in a sequence of image subframes, such as three or

more subframes. In each subframe, one or more of the light sources **201**, **202** and **203** may be projected to provide an illumination pattern onto the PLM **210** and accordingly the SLM **220**. The illumination projections **500** include a first illumination projection **510** of the light beams from the light sources **201** and **202** on the PLM **210**. The first illumination projection **510** includes first illumination patterns **511** and **512** of the light sources **201** and **202**, respectively, without the light source **203** which may be switched off during a first subframe **519**. The first illumination patterns **511** and **512** may have uniform and nonoverlapping illumination areas on the PLM **210**. The first illumination patterns **511** and **512** may have a first arrangement on the surface of the PLM **210**. The PLM **210** may be controlled according to a first voltage setting to form a first hologram **515**, which reflects the light beams from the PLM **210** providing, in the first subframe **519**, a second illumination projection **520** on the SLM **220**. The second illumination projection **520** includes second illumination patterns **521** and **522** of the light beams from the respective light sources **201** and **202**. The second illumination patterns **521** and **522** may have uniform and nonoverlapping illumination areas on the SLM **220**. The second illumination patterns **521** and **522** may be projected on respective pixels or pixel portions **524** and **525** at the SLM **220**. According to the first hologram **515**, the second illumination patterns **521** and **522** may be projected to match the first arrangement of the first illumination patterns **511** and **512** on the surface of the PLM **210**.

To scroll the light beams of the light sources **201** and **202** on the SLM **220**, the PLM **210** may be controlled according to a second voltage setting to form a second hologram **527**, which provides in a second subframe **529** a third illumination projection **530** of the light beams on the SLM **220**. The second hologram **527** causes the light beams of the first illumination patterns **511** and **512** to be reflected from the PLM **210** to the SLM **220** at different angles than the first hologram **515** thereby scrolling the light beams on the surface of the SLM **220**. The third illumination projection **530** includes third illumination patterns **531** and **532** of the light beams of the respective light sources **201** and **202**. The third illumination patterns **531** and **532** may have uniform and nonoverlapping illumination areas on the SLM **220**. According to the second hologram **527**, the third illumination patterns **531** and **532** may be projected on the SLM **220** in a second arrangement different than the first arrangement of the first illumination patterns **511** and **512** on the surface of the PLM **210**. According to the second arrangement, the third illumination patterns **531** and **532** may be projected on the pixels or pixel portions **525** and **524**, respectively.

The illumination projections **500** also include a fourth illumination projection **540** of the light beams from the light source **203** on the PLM **210**. The fourth illumination projection **540** may be provided on the surface of the PLM **210** in a third subframe **549** after the first illumination projection **510**. The fourth illumination projection **540** includes a single illumination pattern **513** of the light source **203** without the light sources **201** and **202**, which may be switched off. The PLM **210** may be controlled according to a third voltage setting to form a third hologram **545**, which reflects the light beam of the light source **203** from the PLM **210** providing in the third subframe **549** a fifth illumination projection **550** on the SLM **220**. The fifth illumination projection **550** includes a single fifth illumination pattern **553** of the light source **203**. The fifth illumination pattern **553** may provide a single color mode of the light source **203** projected on both the pixels or pixel portions **524** and **525** at the SLM **220**. For example, the single color mode may be for blue light. The

second illumination projection **520**, third illumination projection **530**, and fifth illumination projection **550** may be switched on the SLM **220** at a rate allowing the human eye to superimpose the second illumination patterns **521** and **522**, third illumination patterns **531** and **532**, and fifth illumination pattern **553** in the same pixels or pixel portions **524** and **525** to perceive a single image. The illumination patterns may be combined with additional illumination patterns in subsequent subframes to project a complete image.

FIG. 6 is a block diagram of illumination projections **600** of an HDR image pattern **601** in the display device **200**, in accordance with various examples. The illumination projections **600** may provide HDR images having brighter illumination and higher contrast in comparison to the images provided by the illumination projections **500** without HDR modulation. For example, the HDR image pattern **601** may be provided by an HDR modulation technique performed while controlling the light beams by the PLM **210**, SLM **220**, or both. The illumination projections **600** include a first illumination projection **610** of the light beams from the light sources **201** and **202** on the PLM **210**. The first illumination projection **610** includes first illumination patterns **611** and **612** of the light sources **201** and **202**, respectively, without the light source **203** which may be switched off. The first illumination patterns **611** and **612** may have uniform and nonoverlapping illumination areas on the PLM **210**. The first illumination patterns **611** and **612** may be projected on the PLM **210** in a first arrangement. The PLM **210** may be controlled according to a first voltage setting to form a first hologram **615**, which reflects the light beams from the PLM **210** providing, in a first subframe **619**, a second illumination projection **620** on the SLM **220**. The second illumination projection **620** includes second illumination patterns **621** and **622** of the light beams from the respective light sources **201** and **202**. The second illumination patterns **621** and **622** may have uniform and nonoverlapping illumination areas on the SLM **220**. The second illumination patterns **621** and **622** may be projected on respective pixels or pixel portions **624** and **625** at the SLM **220**. The second illumination patterns **621** and **622** may include HDR level contrast in each pixel portion **624** and **625** where the image portions may be brighter than and illuminated without non-image portions in each pixel according to the HDR image pattern **601**. According to the first hologram **615**, the color distribution of the second illumination patterns **621** and **622** may be projected to match the color distribution of the first illumination patterns **611** and **612** on the PLM **210**. The spatial distribution of the second illumination patterns **621** and **622** may match the HDR image pattern **601**.

To scroll the light beams on the SLM **220**, the PLM **210** may be controlled according to a second voltage setting to form a second hologram **627**, which provides in a second subframe **629** a third illumination projection **630** of the light beams on the SLM **220**. The second hologram **627** causes the light beams of the first illumination patterns **611** and **612** to be reflected from the PLM **210** to the SLM **220** at different angles than the first hologram **615** thereby scrolling the light beams on the surface of the SLM **220**. The third illumination projection **630** includes third illumination patterns **631** and **632** of the light beams of the respective light sources **201** and **202**. The third illumination patterns **631** and **632** may have uniform and nonoverlapping illumination areas on the SLM **220**. According to the second hologram **627**, the third illumination patterns **631** and **632** may be projected in a second arrangement different than the first arrangement of the first illumination patterns **611** and **612** on the PLM **210**.

According to the arrangement, the third illumination patterns **631** and **632** may be projected on the pixels or pixel portions **625** and **624**, respectively. The third illumination patterns **631** and **632** may include HDR level contrast in each pixel portion **625** and **624** where the image portions may be brighter than and illuminated without non-image portions in each pixel according to the HDR image pattern **601**.

The illumination projections **600** also include a fourth illumination projection **640** of the light beams from the light source **203** on the PLM **210**. The fourth illumination projection **640** may be provided on the surface of the PLM **210** in a third subframe **649** after the first illumination projection **610**. The fourth illumination projection **640** includes a single illumination pattern **613** of the light source **203** without the light sources **201** and **202**, which may be switched off. The PLM **210** may be controlled according to a third voltage setting to form a third hologram **645**, which reflects the light beam of the light source **203** from the PLM **210** providing in the third subframe **649** a fifth illumination projection **650** on the SLM **220**. The fifth illumination projection **650** includes a single fifth illumination pattern **653** of the light source **203**. The fifth illumination pattern **653** may provide a single color mode of the light source **203** projected on both the pixels or pixel portions **624** and **625** at the SLM **220**. For example, the single color mode may be for blue light. The second illumination projection **620**, third illumination projection **630**, and fifth illumination projection **650** may be switched on the SLM **220** at a rate allowing the human eye to superimpose the second illumination patterns **621** and **622**, third illumination patterns **631** and **632**, and fifth illumination pattern **653** in the same pixels or pixel portions **624** and **625** to perceive a single image according to the HDR image pattern **601**. The illumination patterns may be combined with additional illumination patterns in subsequent subframes to project a complete image.

FIG. 7 is a block diagram of a display device **700** which may be part of the display projector **110**, in accordance with various examples. The display device **700** may have a different arrangement of optical components than the display device **200**. In FIG. 7, the display device **700** may include light sources **701**, **702**, and **703**, a PLM **710**, an SLM **720**, and a projection lens **730**. The light sources **701**, **702**, and **703** may be any light emitting devices that emit respective light beams, in different wavelengths of different color modes, directed toward the PLM **710**. The display device **700** may also include lenses **741**, **742**, and **743** positioned between the light sources **701**, **702**, and **703**, respectively, and the PLM **710**. Each one of the light sources **701**, **702**, and **703** may be aligned with the respective lenses **741**, **742**, and **743**. The lenses **741**, **742**, and **743** are designed for projecting the respective light beams from the light sources **701**, **702**, and **703** onto a surface of the PLM **710**.

The lens **743** of the light source **703** may have a larger surface area or diameter than the lenses **741** and **742** of the respective light sources **701** and **702**. The lens **743** may have a longer focal length than the lenses **741** and **742** to project a light beam with a larger width than the respective light beams projected by the lenses **741** and **742**. The lenses **741** and **742** may have approximately the same surface area and may be positioned at respective distances and locations from the respective light sources **701** and **702** to provide uniform and nonoverlapping illumination patterns of the light beams on the surface of the PLM **710**. The lenses **741** and **742** may have approximately the same focal length or different focal lengths based on the properties of the respective light sources **701** and **702**, such as the divergence angles and the

spatial dimensions of the respective light beams. The lens **743** may be positioned at a distance and location from the light source **703** to provide a larger illumination pattern for the light source **703** on the surface of the PLM **710** than the illumination patterns of the light sources **701** and **702** projected by the lenses **741** and **742**. The lens **743** may be useful to provide a larger illumination area of the color mode of the light source **703** than the color modes of the light sources **701** and **702**. For example, the illumination area of the light source **703** may overlap with both the illuminations areas of the light sources **701** and **702**. The color mode of the light source **703** may be for blue light and the color modes of the light sources **701** and **702** may be for red light and green light.

The display device **700** may include a reflector **750** between the lenses **741**, **742**, and **743** and the PLM **710**. The reflector **750** may be useful for projecting parallel light beams from the light sources **701**, **702**, and **703** on the surface of the PLM **710**. For example, the reflector **750** may have a reflective coating that acts as a filter on a surface of the reflector **750** facing the light source **703** at an approximately 45 degrees (°) angle to reflect the light beam from the light source **703** at approximately 90° from the light source **703** to the PLM **710**. For example, the coating may be designed to reflect blue light and transmit green and red light. The reflector **750** may also have a transparent surface facing the light sources **701** and **702** at approximately 45° to allow straight and parallel light beams to pass through from the light sources **701** and **702** to the PLM **710**. In other examples, the angle between the reflector **750** and the light sources **701**, **702**, and **703** may be any suitable angle to project parallel light beams from the light sources **701**, **702**, and **703** onto the surface of the PLM **710**. In examples, the reflector **750** may be a dichroic mirror or filter having different reflection and transmission properties for different wavelengths. The dichroic mirror or filter may be a prism that allows light beams, at wavelengths of the respective color modes of the light sources **701** and **702**, to pass onto the PLM **710** and reflects a light beam, at a wavelength of the color mode of the light source **703**, at approximately 90° onto the PLM **710**. In other examples, the reflector **750** may be a polarizing beam splitter (PBS) configured to split light into two light beams of two respective light polarization states. The PBS may be a cube that allows light beams, polarized in a first polarization state, to pass through from the light sources **701** and **702** to the PLM **710** and that reflects a light beam, polarized in a second polarization state, at approximately 90° from the light source **703** to the PLM **710**.

FIG. 8 is a block diagram of illumination projections **800** in the display device **700**, in accordance with various examples. The illumination projections **800** may be projected in a sequence of image subframes, such as three or more subframes. In each subframe, one or more of the light sources **701**, **702** and **703** may be projected to provide an illumination pattern onto the PLM **710** and accordingly the SLM **720**. The light beam of at least one of the light sources **701**, **702** and **703** may cover a different size illumination area on the surface of the PLM **710** in comparison to the other light sources. For example, the light source **703** may project through the lens **743** a larger light beam that covers a larger area on the surface of the PLM **710** than the light sources **701** and **702**. The illumination projections **800** include a first illumination projection **810** of the light beams from the light sources **701** and **702** on the PLM **710**. The first illumination projection **810** includes first illumination patterns **811** and **812** of the light sources **701** and **702**, respec-

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tively, without the light source **703** which may be switched off. The first illumination patterns **811** and **812** may have uniform and nonoverlapping illumination areas on the PLM **710**. The first illumination patterns **811** and **812** may be projected in a first arrangement on the PLM **710**. The PLM **710** may be controlled according to a first voltage setting to form a first hologram **815**, which reflects the light beams from the PLM **710** providing, in a first subframe **819**, a second illumination projection **820** of the light beams on the SLM **720**. The second illumination projection **820** includes second illumination patterns **821** and **822** of the light beams from the respective light sources **701** and **702**. The second illumination patterns **821** and **822** may have uniform and nonoverlapping illumination areas on the SLM **720**. The second illumination patterns **821** and **822** may be projected on respective pixels or pixel portions **824** and **825** at the SLM **720**. According to the first hologram **815**, the second illumination patterns **821** and **822** may be projected to match the first arrangement of the first illumination patterns **811** and **812** on the PLM **710**.

To scroll the light beams on the SLM **720**, the PLM **710** may be controlled according to a second voltage setting to form a second hologram **827**, which provides in a second subframe **829** a third illumination projection **830** of the light beams on the SLM **720**. The second hologram **827** causes the light beams of the first illumination patterns **811** and **812** to be reflected from the PLM **710** to the SLM **720** at different angles than the first hologram **815** thereby scrolling the light beams on the surface of the SLM **720**. The third illumination projection **830** includes third illumination patterns **831** and **832** of the light beams of the respective light sources **701** and **702**. The third illumination patterns **831** and **832** may have uniform and nonoverlapping illumination areas on the SLM **720**. According to the second hologram **827**, the third illumination patterns **831** and **832** may be projected on the SLM **720** in a second arrangement different than the first arrangement of the first illumination patterns **811** and **812**. According to the second arrangement, the third illumination patterns **831** and **832** may be projected on the pixels or pixel portions **825** and **824**, respectively.

The illumination projections **800** also include a fourth illumination projection **840** of the light beams from the light source **703** on the PLM **710**. The fourth illumination projection **840** may be provided on the surface of the PLM **710** in a third subframe **849** after the first illumination projection **810**. The fourth illumination projection **840** includes a single fourth illumination pattern **813** of the light source **703** without the light sources **701** and **702**, which may be switched off. The fourth illumination pattern **813** may cover a larger illumination area on the surface of the PLM **710** with respect to the first illumination patterns **811** and **812**. The illumination area of the fourth illumination pattern **813** on the surface of the PLM **710** may be proportional to the larger surface area of the lens **743** with respect to the lenses **741** and **742**. For example, the illumination area of the fourth illumination pattern **813** may be approximately equivalent to the combined illumination area of the first illumination patterns **811** and **812**. The PLM **710** may be controlled according to a third voltage setting to form a third hologram **845**, which reflects the light beams from the PLM **710** providing, in the third subframe **849**, a fifth illumination projection **850** on the SLM **720**. The fifth illumination projection **850** includes a single fifth illumination pattern **853** of the light source **703**. The fifth illumination pattern **853** may provide a single color mode of the light source **703** projected on both the pixels or pixel portions **824** and **825** at the SLM **720**. For example, the single color mode may be for

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blue light. The second illumination projection **820**, third illumination projection **830**, and fifth illumination projection **850** may be switched on the SLM **720** at a speed allowing the human eye to superimpose the second illumination patterns **821** and **822**, third illumination patterns **831** and **832**, and fifth illumination pattern **853** in the same pixels or pixel portions **824** and **825** to perceive a single image. The illumination patterns may be combined with additional illumination patterns in subsequent subframes to project a complete image.

FIG. **9** is a block diagram of illumination projections **900** of an HDR image pattern **901** in the display device **700**, in accordance with various examples. The illumination projections **900** may provide HDR images having brighter illumination and higher contrast in comparison to the images provided by the illumination projections **900** without HDR modulation. For example, the HDR image pattern **901** may be provided by an HDR modulation technique performed while controlling the light beams by the PLM **710**, SLM **720**, or both. The illumination projections **900** include a first illumination projection **910** of the light beams from the light sources **701** and **702** on the PLM **710**. The first illumination projection **910** includes first illumination patterns **911** and **912** of the light sources **701** and **702**, respectively, without the light source **703** which may be switched off. The first illumination patterns **911** and **912** may have uniform and nonoverlapping illumination areas on the PLM **710**. The first illumination patterns **911** and **912** may be projected on the PLM **710** in a first arrangement. The PLM **710** may be controlled according to a first voltage setting to form a first hologram **915**, which reflects the light beams from the PLM **710** providing, in a first subframe **919**, a second illumination projection **920** on the SLM **720**. The second illumination projection **920** includes second illumination patterns **921** and **922** of the light beams from the respective light sources **701** and **702**. The second illumination patterns **921** and **922** may have uniform and nonoverlapping illumination areas on the SLM **720**. The second illumination patterns **921** and **922** may be projected on the respective pixels or pixel portions **924** and **925** at the SLM **720**. The second illumination patterns **921** and **922** may include HDR level contrast in each pixel portion **924** and **925** where the image portions may be brighter than and illuminated without non-image portions in each pixel according to the HDR image pattern **901**. According to the first hologram **915**, the second illumination patterns **921** and **922** may be projected to match the color distribution of the first illumination patterns **911** and **912** on the PLM **710**. The spatial distribution of the second illumination patterns **921** and **922** may match the HDR image pattern **901**.

To scroll the light beams on the SLM **720**, the PLM **710** may be controlled according to a second voltage setting to form a second hologram **927**, which provides in a second subframe **929** a third illumination projection **930** of the light beams on the SLM **720**. The second hologram **927** causes the light beams of the first illumination patterns **911** and **912** to be reflected from the PLM **710** to the SLM **720** at different angles than the first hologram **915** thereby scrolling the light beams on the surface of the SLM **720**. The third illumination projection **930** includes third illumination patterns **931** and **932** of the light beams of the respective light sources **701** and **702**. The third illumination patterns **931** and **932** may have uniform and nonoverlapping illumination areas on the SLM **720**. According to the second hologram **927**, the third illumination patterns **931** and **932** may be projected on the SLM **720** in a second arrangement different than the first arrangement of the first illumination patterns **911** and **912**.

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According to the second arrangement, the third illumination patterns **931** and **932** may be projected on the pixels or pixel portions **925** and **924**, respectively. The third illumination patterns **931** and **932** may include HDR level contrast in each pixel portion **925** and **924** where the image portions may be brighter than and illuminated without non-image portions in each pixel according to the HDR image pattern **901**.

The illumination projections **900** also include a fourth illumination projection **940** of the light beams from the light source **703** on the PLM **710**. The fourth illumination projection **940** may be provided on the surface of the PLM **710** in a third subframe **949** after the first illumination projection **910**. The fourth illumination projection **940** may include a single fourth illumination pattern **913** of the light source **703** without the light sources **701** and **702**, which may be switched off. The fourth illumination pattern **913** may cover a larger illumination area on the surface of the PLM **710** with respect to the first illumination patterns **911** and **912**. The illumination area of the fourth illumination pattern **913** on the surface of the PLM **710** may be proportional to the larger surface area of the lens **743** with respect to the lenses **741** and **742**. For example, the illumination area of the fourth illumination pattern **913** may be approximately equivalent to the combined illumination area of the first illumination patterns **911** and **912**. The PLM **710** may be controlled according to a third voltage setting to form a third hologram **945**, which reflects the light beam of the light source **703** from the PLM **710** providing, in the third subframe **949**, a fifth illumination projection **950** on the SLM **720**. The fifth illumination projection **950** includes a single fifth illumination pattern **953** of the light source **703**. The fifth illumination pattern **953** may provide a single color mode of the light source **703** projected on both the pixels or pixel portions **924** and **925** at the SLM **720**. For example, the single color mode may be for blue light. The second illumination projection **920**, third illumination projection **930**, and fifth illumination projection **950** may be switched on the SLM **720** at a rate allowing the human eye to superimpose the second illumination patterns **921** and **922**, third illumination patterns **931** and **932**, and fifth illumination pattern **953** in the same pixels or pixel portions **924** and **925** to perceive a single image according to the HDR image pattern **901**. The illumination patterns may be combined with additional illumination patterns in subsequent subframes to project a complete image.

FIG. **10** is a block diagram of illumination projections **1000** in the display device **700**, in accordance with various examples. The illumination projections **1000** may be projected in a sequence of image subframes, such as three or more subframes. In each subframe, one or more of the light sources **701**, **702** and **703** may be projected to provide an illumination pattern onto the PLM **710** and accordingly the SLM **720**. The light beam of at least one of the light sources **701**, **702** and **703** may cover a different size illumination area on the surface of the PLM **710** in comparison to the other light sources. For example, the light source **703** may project through the lens **743** a larger light beam that covers a larger area on the surface of the PLM **710** than the light sources **701** and **702**. Two or more of the light sources **701**, **702** and **703** may also be projected onto the same surface area of the PLM **710** to provide a new color mode. The new color mode may be a combination of the color modes of the light beams superimposed on the same surface area of the PLM **710**. The illumination projections **1000** include a first illumination projection **1010** of the light beams from the light sources **701** and **702** on the PLM **710**. The first

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illumination projection **1010** includes first illumination patterns **1011** and **1012** of the light sources **701** and **702**, respectively, without the light source **703** which may be switched off. The first illumination patterns **1011** and **1012** may have uniform and nonoverlapping illumination areas on the PLM **710**. The first illumination patterns **1011** and **1012** may be projected on the PLM **710** in a first arrangement. The PLM **710** may be controlled according to a first voltage setting to form a first hologram **1015**, which reflects the light beams from the PLM **710** providing, in a first subframe **1019**, a second illumination projection **1020** on the SLM **720**. The second illumination projection **1020** includes second illumination patterns **1021** and **1022** of the light beams from the respective light sources **701** and **702**. The second illumination patterns **1021** and **1022** may have uniform and nonoverlapping illumination areas on the SLM **720**. The second illumination patterns **1021** and **1022** may be projected on respective pixels or pixel portions **1024** and **1025** at the SLM **720**. According to the first hologram **1015**, the second illumination patterns **1021** and **1022** may be projected to match the first arrangement of the first illumination patterns **1011** and **1012** on the PLM **710**.

To scroll the light beams on the SLM **720**, the PLM **710** may be controlled according to a second voltage setting to form a second hologram **1027**, which provides in a second subframe **1029** a third illumination projection **1030** of the light beams on the SLM **720**. The second hologram **1027** causes the light beams of the first illumination patterns **1011** and **1012** to be reflected from the PLM **710** to the SLM **720** at different angles than the first hologram **1015** thereby scrolling the light beams on the surface of the SLM **720**. The third illumination projection **1030** includes third illumination patterns **1031** and **1032** of the light beams of the respective light sources **701** and **702**. The third illumination patterns **1031** and **1032** may have uniform and nonoverlapping illumination areas on the SLM **720**. According to the second hologram **1027**, the third illumination patterns **1031** and **1032** may be projected on the SLM **720** in a second arrangement different than the first arrangement of the first illumination patterns **1011** and **1012**. According to the second arrangement, the third illumination patterns **1031** and **1032** may be projected on the pixels or pixel portions **1025** and **1024**, respectively.

The illumination projections **1000** include a fourth illumination projection **1040** of the light beams from the light source **703** on the PLM **710**. The fourth illumination projection **1040** may be provided on the surface of the PLM **710** in a third subframe **1049** after the first illumination projection **1010**. The fourth illumination projection **1040** includes a single fourth illumination pattern **1013** of the light source **703** without the light sources **701** and **702**, which may be switched off. The fourth illumination pattern **1013** may cover a larger illumination area on the surface of the PLM **710** with respect to the first illumination patterns **1011** and **1012**. The illumination area of the fourth illumination pattern **1013** on the surface of the PLM **710** may be proportional to the larger surface area of the lens **743** with respect to the lenses **741** and **742**. For example, the illumination area of the fourth illumination pattern **1013** may be approximately equivalent to the combined illumination area of the first illumination patterns **1011** and **1012**. The PLM **710** may be controlled according to a third voltage setting to form a third hologram **1045**, which reflects the light beam of the light source **703** from the PLM **710** providing, in the third subframe **1049**, a fifth illumination projection **1050** on the SLM **720**. The fifth illumination projection **1050** includes a single fifth illumination pattern **1053** of the light source **703**.

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The fifth illumination pattern **1053** may provide a single color mode of the light source **703** projected on both the pixels or pixel portions **1024** and **1025** at the SLM **720**. For example, the single color mode may be for blue light.

The illumination projections **1000** also includes a sixth illumination projection **1060** with overlapping light beams from the light sources **701**, **702**, and **703** on the SLM **720**. The sixth illumination projection **1060** includes overlapping illumination patterns **1054** and **1055** from the light sources **701**, **702**, and **703** in a fourth arrangement on the PLM **710**. The sixth illumination projection **1060** may be provided in a fourth subframe **1069** as a combination of the first illumination projection **1010** and the fourth illumination projection **1040**. The sixth illumination projection **1060** is a superposition of the first illumination patterns **1011** and **1012** and the single fourth illumination pattern **1013** on the PLM **710**. For example, the first illumination patterns **1011** and **1012** may be uniform and nonoverlapping illumination areas of red light and green light, respectively, and the single fourth illumination pattern **1013** may be an illumination area of blue light that overlaps with the nonoverlapping illumination areas of red light and green light. Accordingly, the resulting overlapping illumination patterns **1054** and **1055** may be uniform and nonoverlapping illumination areas of magenta light and cyan light, respectively, on the PLM **710**. The overlapping illumination patterns **1054** and **1055** may be projected on the PLM **710** in a third arrangement. The PLM **710** may be controlled according to a fourth voltage setting to form a fourth hologram **1065**, which reflects the light beams from the PLM **710** providing, in the fourth subframe **1069**, a seventh illumination projection **1070** of the light beams on the SLM **720**. The seventh illumination projection **1070** includes second overlapping illumination patterns **1071** and **1072** of the same color modes of the overlapping illumination patterns **1054** and **1055**, respectively. The second overlapping illumination patterns **1071** and **1072** may have uniform and nonoverlapping illumination areas on the SLM **720**. The second overlapping illumination patterns **1071** and **1072** may be projected on respective pixels or pixel portions **1024** and **1025** at the SLM **720**. According to the fourth hologram **1065**, the second overlapping illumination patterns **1071** and **1072** may be projected to match the first arrangement of the overlapping illumination patterns **1054** and **1055** on the PLM **710**.

To scroll the light beams on the SLM **720**, the PLM **710** may be controlled according to a fifth voltage setting to form a fifth hologram **1085**, which provides in a fifth subframe **1089** an eighth illumination projection **1090** of the light beams on the SLM **720**. The fifth hologram **1085** causes the light beams of the sixth illumination projection **1060** to be reflected from the PLM **710** to the SLM **720** at different angles than the fourth hologram **1065** thereby scrolling the light beams on the surface of the SLM **720**. The eighth illumination projection **1090** includes third overlapping illumination patterns **1091** and **1092** of the same color modes of the overlapping illumination patterns **1054** and **1055**. The third overlapping illumination patterns **1091** and **1092** may have uniform and nonoverlapping illumination areas on the SLM **720**. According to the fifth hologram **1085**, the third overlapping illumination patterns **1091** and **1092** may be projected on the SLM **720** in a fourth arrangement different from the third arrangement of the overlapping illumination patterns **1054** and **1055**. According to the fourth arrangement, the third overlapping illumination patterns **1091** and **1092** may be projected on the pixels or pixel portions **1025** and **1024** respectively.

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The second illumination projection **1020**, third illumination projection **1030**, fifth illumination projection **1050**, seventh illumination pattern **1070**, and eighth illumination projection **1090** may be switched on the SLM **720** at a rate allowing the human eye to superimpose the second illumination patterns **1021** and **1022**, third illumination patterns **1031** and **1032**, fifth illumination pattern **1053**, second overlapping illumination patterns **1071** and **1072**, and third overlapping illumination patterns **1091** and **1092** in the same pixels or pixel portions **1024** and **1025** to perceive a single image. The illumination patterns may be combined with additional illumination patterns in subsequent subframes to project a complete image.

FIG. **11** is a block diagram of illumination projections **1100** of an HDR image pattern **1101** in the display device **700**, in accordance with various examples. The illumination projections **1100** may provide HDR images having brighter illumination and higher contrast in comparison to the images provided by the illumination projections **1100** without HDR modulation. For example, the HDR image pattern **1101** may be provided by an HDR modulation technique performed while controlling the light beams by the PLM **710**, SLM **720**, or both. The illumination projections **1100** include a first illumination projection **1110** of the light beams from the light sources **701** and **702** on the PLM **710**. The first illumination projection **1110** includes first illumination patterns **1111** and **1112** of the light sources **701** and **702**, respectively, without the light source **703** which may be switched off. The first illumination patterns **1111** and **1112** may have uniform and nonoverlapping illumination areas on the PLM **710**. The first illumination patterns **1111** and **1112** may be projected on the PLM **710** in a first arrangement. The PLM **710** may be controlled according to a first voltage setting to form a first hologram **1115**, which reflects the light beams from the PLM **710** providing, in a first subframe **1119**, a second illumination projection **1120** on the SLM **720**. The second illumination projection **1120** includes second illumination patterns **1121** and **1122** of the light beams from the respective light sources **701** and **702**. The second illumination patterns **1121** and **1122** may have uniform and nonoverlapping illumination areas on the SLM **720**. The second illumination patterns **1121** and **1122** may be projected on respective pixels or pixel portions **1124** and **1125** at the SLM **720**. The second illumination patterns **1121** and **1122** may include HDR level contrast in each pixel portion **1124** and **1125** where the image portions may be brighter than and illuminated without non-image portions in each pixel according to the HDR image pattern **1101**. According to the first hologram **1115**, the second illumination patterns **1121** and **1122** may be projected to match the color distribution of the first illumination patterns **1111** and **1112** on the PLM **710**. The spatial distribution of the second illumination patterns **1121** and **1122** may match the HDR image pattern **1101**.

To scroll the light beams on the SLM **720**, the PLM **710** may be controlled according to a second voltage setting to form a second hologram **1127**, which provides in a second subframe **1129** a third illumination projection **1130** of the light beams on the SLM **720**. The second hologram **1127** causes the light beams of the first illumination patterns **1111** and **1112** to be reflected from the PLM **710** to the SLM **720** at different angles than the first hologram **1115** thereby scrolling the light beams on the surface of the SLM **720**. The third illumination projection **1130** includes third illumination patterns **1131** and **1132** of the light beams of the respective light sources **701** and **702**. The third illumination patterns **1131** and **1132** may have uniform and nonoverlapping

ping illumination areas on the SLM 720. According to the second hologram 1127, the third illumination patterns 1131 and 1132 may be projected on the SLM 720 in a second arrangement different than the first arrangement of the first illumination patterns 1111 and 1112. According to the second arrangement, the third illumination patterns 1131 and 1132 may be projected on the pixels or pixel portions 1125 and 1124, respectively. The third illumination patterns 1131 and 1132 may include HDR level contrast in each pixel portion 1125 and 1124 where the image portions may be brighter than and illuminated without non-image portions in each pixel according to the HDR image pattern 1101.

The illumination projections 1100 include a fourth illumination projection 1140 of the light beams from the light source 703 on the PLM 710. The fourth illumination projection 1140 may be provided on the surface of the PLM 710 in a third subframe 1149 after the first illumination projection 1110. The fourth illumination projection 1140 includes a single fourth illumination pattern 1113 of the light source 703 without the light sources 701 and 702, which may be switched off. The fourth illumination pattern 1113 may cover a larger illumination area on the surface of the PLM 710 with respect to the first illumination patterns 1111 and 1112. The illumination area of the fourth illumination pattern 1113 on the surface of the PLM 710 may be proportional to the larger surface area of the lens 743 with respect to the lenses 741 and 742. For example, the illumination area of the fourth illumination pattern 1113 may be approximately equivalent to the combined illumination area of the first illumination patterns 1111 and 1112. The PLM 710 may be controlled according to a third voltage setting to form a third hologram 1145, which reflects the light beam of the light source 703 from the PLM 710 providing, in the third subframe 1149, a fifth illumination projection 1150 on the SLM 720. The fifth illumination projection 1150 includes a single fifth illumination pattern 1153 of the light source 703. The fifth illumination pattern 1153 may provide a single color mode of the light source 703 projected on both the pixels or pixel portions 1124 and 1125 at the SLM 720. For example, the single color mode may be for blue light.

The illumination projections 1100 also includes a sixth illumination projection 1160 with overlapping light beams from the light sources 701, 702, and 703 on the SLM 720. The sixth illumination projection 1160 includes overlapping illumination patterns 1154 and 1155 from the light sources 701, 702, and 703 in a fourth arrangement on the PLM 710. The sixth illumination projection 1160 may be provided in a fourth subframe 1169 as a combination of the first illumination projection 1110 and the fourth illumination projection 1140. The sixth illumination projection 1160 is a superposition of the first illumination patterns 1111 and 1112 and the single fourth illumination pattern 1113 on the PLM 710. For example, the first illumination patterns 1111 and 1112 may be uniform and nonoverlapping illumination areas of red light and green light, respectively, and the single fourth illumination pattern 1113 may be an illumination area of blue light that overlaps with the nonoverlapping illumination areas of red light and green light. Accordingly, the resulting overlapping illumination patterns 1154 and 1155 may be uniform and nonoverlapping illumination areas of magenta light and cyan light, respectively, on the PLM 710. The overlapping illumination patterns 1154 and 1155 may be projected on the PLM 710 in a third arrangement. The PLM 710 may be controlled according to a fourth voltage setting to form a fourth hologram 1165, which reflects the light beams from the PLM 710 providing, in the fourth subframe 1169, a seventh illumination projection 1170 of the light

beams on the SLM 720. The seventh illumination projection 1170 includes second overlapping illumination patterns 1171 and 1172 of the same color modes of the overlapping illumination patterns 1154 and 1155, respectively. The second overlapping illumination patterns 1171 and 1172 may have uniform and nonoverlapping illumination areas on the SLM 720. The second overlapping illumination patterns 1171 and 1172 may be projected on respective pixels or pixel portions 1124 and 1125 at the SLM 720. According to the fourth hologram 1165, the second overlapping illumination patterns 1171 and 1172 may be projected to match the first arrangement of the overlapping illumination patterns 1154 and 1155 on the PLM 710.

To scroll the light beams on the SLM 720, the PLM 710 may be controlled according to a fifth voltage setting to form a fifth hologram 1185, which provides in a fifth subframe 1189 an eighth illumination projection 1190 of the light beams on the SLM 720. The fifth hologram 1185 causes the light beams of the sixth illumination projection 1160 to be reflected from the PLM 710 to the SLM 720 at different angles than the fourth hologram 1165 thereby scrolling the light beams on the surface of the SLM 720. The eighth illumination projection 1190 includes third overlapping illumination patterns 1191 and 1192 of the same color modes of the overlapping illumination patterns 1154 and 1155. The third overlapping illumination patterns 1191 and 1192 may have uniform and nonoverlapping illumination areas on the SLM 720. According to the fifth hologram 1185, the third overlapping illumination patterns 1191 and 1192 may be projected on the SLM 720 in a fourth arrangement different from the third arrangement of the overlapping illumination patterns 1154 and 1155. According to the fourth arrangement, the third overlapping illumination patterns 1191 and 1192 may be projected on the pixels or pixel portions 1125 and 1124 respectively.

The second illumination projection 1120, third illumination projection 1130, fifth illumination projection 1150, seventh illumination pattern 1170, and eighth illumination projection 1190 may be switched on the SLM 720 at a rate allowing the human eye to superimpose the second illumination patterns 1121 and 1122, third illumination patterns 1131 and 1132, fifth illumination pattern 1153, second overlapping illumination patterns 1171 and 1172, and third overlapping illumination patterns 1191 and 1192 in the same pixels or pixel portions 1124 and 1125 to perceive a single image according to the HDR image pattern 1101. The illumination patterns may be combined with additional illumination patterns in subsequent subframes to project a complete image.

FIG. 12 is a block diagram of a display device 1200 which may be part of the display projector 110, in accordance with various examples. The display device 1200 may have a different arrangement of optical components than the display device 200 or the display device 700. In FIG. 12, the display device 1200 may include light sources 1201, 1202, and 1203, a PLM 1210, an SLM 1220, and a projection lens 1230. The light sources 1201, 1202, and 1203 may be any light emitting devices that emit respective light beams, in different wavelengths of different color modes, directed toward the PLM 1210. The display device 1200 may also include lenses 1240 positioned between the light sources 1201, 1202, and 1203, respectively, and the PLM 1210. Each one of the light sources 1201, 1202, and 1203 may be aligned with the respective lenses 1240. The lenses 1240 are designed for projecting the respective light beams from the light sources 1201, 1202, and 1203 onto a surface of the PLM 1210. The light beams may be reflected from the PLM

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1210 onto the SLM 1220 without obstruction. In the example shown in FIG. 12, the light source 1203 may be placed behind the PLM 1210 and the SLM 1220 and behind the line of projection of the light beams between the PLM 1210 and the SLM 1220. In other examples, the light sources 1201, 1202 and 1203 may be positioned in any suitable arrangement that does not obstruct the lines of projections of the light beams between the components of the display device 1200.

The lenses 1240 may have approximately the same surface and may be positioned at respective distances and locations from the respective light sources 1201, 1202, and 1203 to provide uniform and nonoverlapping illumination patterns of the light beams on the surface of the PLM 1210. The display device 1200 may include one or more reflectors 1250 positioned between the PLM 1210 and one or more of the lenses 1240 and the PLM 1210. The reflectors 1250 may be mirrors that direct respective light beams from the one or more of the lenses 1240 to the PLM 1210. The reflectors 1250 may be useful for projecting parallel light beams from the one or more of the lenses 1240 on the surface of the PLM 710 and providing a compact arrangement of light sources 1201, 1202, and 1203 in the display projector 110.

For example, as shown in FIG. 12, a first reflector 1250 may be positioned between the light source 1201 and the PLM 1210. The reflector 1250 may have a reflective surface facing the light source 1201 at an approximately 45° angle to reflect the light beam from the light source 1201 at approximately 90° from the light source 1201 to the PLM 1210. Similarly, a second reflector 1250 may be positioned between the light source 1203 and the PLM 1210. The second reflector 1250 may also have a reflective surface facing the light source 1203 at an approximately 45° angle to reflect the light beam from the light source 1203 at approximately 90° from the light source 1203 to the PLM 1210. The light source 1202 may be facing the PLM 1210 to transmit a straight and undeflected light beam through a respective lens 1240 to the PLM 710. In other examples, the angles between the reflectors 1250 and the one or more of the light sources 1201, 1202, and 1203 may be any suitable angle to project parallel light beams from the one or more of the light sources 1201, 1202, and 1203 onto the surface of the PLM 710.

FIG. 13 is a block diagram of illumination patterns 1300 that may be provided by any of the display devices 200, 700, and 1200, in accordance with various examples. As an example, the display device 200 is shown in FIG. 13. The illumination patterns 1300 may be provided by projecting the light beams from the respective light sources onto the surface of the PLM 210, 710, or 1210. The illumination patterns 1300 may be shaped based on the divergence angles and the spatial dimensions of the respective light beams and the shapes of the lenses in the display device 200, 700, or 1200. For example, responsive to the lenses in the display device 200, 700, or 1200 having similar elliptical shapes 1340, the illumination patterns 1300 may have nonoverlapping elliptical profiles 1350 on the surface of the PLM 210, 710, or 1210. In other examples, the display device 200, 700, or 1200 may include optical components 1380 having different curvatures in the x and y axes, such as cylindrical lenses, prism pairs, rectangular diffusers, light tunnels or rods. Accordingly, the illumination patterns 1300 may have nonoverlapping rectangular profiles 1390 on the surface of the PLM 210, 710, or 1210.

FIG. 14 is a block diagram of a display device 1400 which may be part of the display projector 110, in accordance with various examples. The display device 1400 may include

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multiple SLMs optically coupled to a PLM to scroll light beams from respective light sources. For example, the display device 1400 may include light sources 1401, 1402, and 1403, a PLM 1410, a first SLM 1420, a second SLM 1421, and a projection lens 1430. The light sources 1401, 1402, and 1403 may be any light emitting devices, such as laser light sources, that emit respective light beams in different wavelengths of different color modes. The display device 200 may also include lenses 1440 positioned between the light sources 1401, 1402, and 1403 and the PLM 1410. The PLM 1410 may be configured to reflect the projected light beams from the light sources 1401, 1402, and 1403 onto the first SLM 1420 and the second SLM 1421. Each one of the light beams may be projected by the PLM 1410 onto one of the first SLM 1420 and the second SLM 1421. In the example shown in FIG. 14, the PLM 1410 may reflect the light beams from the light sources 1401 and 1403 onto the first SLM 1420, and reflect the light beam from the light source 1402 onto the second SLM 1421. Accordingly, the PLM 1410 may scroll the light beams from the light sources 1401 and 1403 onto the first SLM 1420 and scroll the light beam from the light source 1402 onto the second SLM 1421. In other examples, a light beam from the same light source may be reflected by the PLM 1410 onto the first SLM 1420 and the second SLM 1421 in different image subframes. For example, the PLM 1410 may reflect the light beam from the light source 1402 onto the second SLM 1421 in a first subframe, and reflect the light beam from the light source 1402 onto the first SLM 1420 in a second subframe.

FIG. 15 is a flow diagram of a method 1500 for light projection in the display device 200, 700, or 1200 of the display projector 110 in the display system 100, in accordance with various examples. At step 1501, light beams at respective wavelengths for respective color modes may be emitted from respective light sources. At least some of the light sources may be turned on continuously during the light projection process. For example, the light sources 201, 202, and 203 may emit respective light beams for the color modes including red, green, and blue to the PLM 210. At step 1502, the light beams from the laser light sources may be projected by respective lenses onto a PLM. The lenses may be useful to project uniform and nonoverlapping illumination patterns of the light beams on the surface of the PLM. For example, the lenses 240 provide similarly shaped and nonoverlapping illumination patterns for the light beams on the surface of the PLM 210.

At step 1503, the light beams may be reflected and scrolled by the PLM simultaneously onto an SLM. The PLM may be controlled with suitable voltages determined with respect to the color modes of the light beams to form holograms on the surface of the PLM. To form the holograms, the light beams with the different wavelengths of different color modes may be reflected with different phase levels on the PLM. Accordingly, the micromirrors of the PLM may be adjusted for different sections of micromirrors on the PLM, which reflect the respective color modes, such as red, green, and blue. This can be performed by applying different voltages to the different sections of micromirrors. The holograms may be changed for each subframe in a sequence of image subframes over a time period to scroll the reflected light beams of the color modes onto the surface of the SLM. The scrolling of the color modes can be quasi-continuous or stepped to discrete positions based on how the image is displayed. For example, the PLM may be controlled to provide holograms with time delays that cause discrete steps in the angles of reflection of the light beams on the SLM and accordingly cause scrolling the light beams

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in discrete steps on the SLM. In other examples, the PLM may be controlled to provide holograms at a rate that causes a quasi-continuous change in the angles of reflection of the light beams on the SLM and accordingly causes scrolling the light beams in quasi-continuous movement on the SLM.

At step 1504, at least portions of the light beams may be reflected by the SLM onto a projection lens. For example, the SLM 220 may reflect the light beams from the PLM 210 onto the projection lens 230 to project the image. The SLM 220 may be controlled by a voltage source to set the states of the micromirrors of the SLM 220 for reflecting portions of the light from the PLM 210 onto the projection lens 230 according to the bright and dark areas of the image. In other examples, the light beams may be reflected and scrolled by the PLM directly onto the projection lens or a display, such as the projection lens 230, without an SLM.

FIG. 16 is a block diagram of a processing system 1600, in accordance with various examples. The processing system 1600 may be part of or coupled to the display system 100 and control components of the display projector 110 to display images or video. For example, the processing system 1600 may be part of the display projector 110 in the display system 100, such as the controller 250 of the display device 200. As shown in FIG. 16, the processing system 1600 may include one or more processors 1601 and one or more memories 1602. In some examples, the processing system 1600 may also include one or more transceivers 1603 and one or more antennas 1604 for establishing wireless connections. These components may be connected through a bus 1605, or in any other suitable manner. In FIG. 16, an example in which the components are connected through a bus 1605 is shown.

The processor 1601 may be configured to read and execute computer-readable instructions. For example, the processor 1601 may be configured to invoke and execute instructions stored in the memory 1602, including the instructions 1606. The processor 1601 may support one or more global systems for wireless communication. Responsive to the processor 1601 sending a message or data, the processor 1601 drives or controls the transceiver 1603 to perform the sending. The processor 1601 also drives or controls the transceiver 1603 to perform receiving, responsive to the processor 1601 receiving a message or data. Therefore, the processor 1601 may be considered as a control center for performing sending or receiving and the transceiver 1603 is an executor for performing the sending and receiving operations.

In some examples, the memory 1602 may be coupled to the processor 1601 through the bus 1605 or an input/output port (not shown). In other examples, the memory 1602 may be integrated with the processor 1601. The memory 1602 is configured to store various software programs and/or multiple groups of instructions, including instructions 1606. For example, the memory 1602 may include a high-speed random-access memory and/or may include a nonvolatile memory such as one or more disk storage devices, a flash memory, or another nonvolatile solid-state storage device. The memory 1602 may store an operating system such as ANDROID, IOS, WINDOWS, or LINUX.

The memory 1602 may further store a network communications program. The network communications program is useful for communication with one or more attached devices, one or more user equipment, or one or more network devices, for example. The memory 1602 may further store a user interface program. The user interface program may display content of an application through a graphical interface and receive a control operation per-

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formed by a user on the application via an input control such as a menu, a dialog box, or a physical input device (not shown). The memory 1602 may be configured to store the instructions 1606 for implementing the various methods and processes provided in accordance with the various examples of this description.

The antenna 1604 may be configured to convert electromagnetic energy into an electromagnetic wave in free space or convert an electromagnetic wave in free space into electromagnetic energy in a transmission line. The transceiver 1603 may be configured to transmit a signal that is provided by the processor 1601 or may be configured to receive a wireless communications signal received by the antenna 1604. In this example, the transceiver 1603 may be considered a wireless transceiver.

The processing system 1600 may also include another communications component such as a Global Positioning System (GPS) module, a BLUETOOTH module, or a WI-FI module. The processing system 1600 may also support another wireless communications signal such as a satellite signal or a short-wave signal. The processing system 1600 may also be provided with a wired network interface or a local area network (LAN) interface to support wired communication.

In various examples, the processing system 1600 may further include an input/output device (not shown) such as an audio input/output device, a key input device, a display, and the like. The input/output device may be configured to implement interaction between the processing system 1600 and a user/an external environment and may include the audio input/output device, the key input device, the display, and the like. The input/output device may further include a camera, a touchscreen, a sensor, and the like. The input/output device may communicate with the processor 1601 through a user interface.

The processing system 1600 shown in FIG. 16 is an example of implementation in various examples of this description. During actual application, the processing system 1600 may include more or fewer components.

The term “couple” appears throughout the specification. The term may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A provides a signal to control device B to perform an action, in a first example device A is coupled to device B, or in a second example device A is coupled to device B through intervening component C if intervening component C does not substantially alter the functional relationship between device A and device B such that device B is controlled by device A via the control signal provided by device A.

A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

A system or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described system or device.

While certain components may be described herein as being of a particular process technology, these components may be exchanged for components of other process technologies. Systems and devices described herein are recon-

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figurable to include the replaced components to provide functionality at least partially similar to functionality available prior to the component replacement.

Unless otherwise stated, “about,” “approximately,” or “substantially” preceding a value means ± 10 percent of the stated value. Modifications are possible in the described examples, and other examples are possible within the scope of the claims.

What is claimed is:

1. An apparatus comprising:
 - a spatial light modulator (SLM) having a first region and a second region; and
 - a phase light modulator (PLM) optically coupled to the SLM, the PLM configured to:
 - receive first light having a first color;
 - receive second light having a second color, the second color different than the first color;
 - direct the first light towards the first region of the SLM; and
 - direct the second light towards the second region of the SLM.
2. The apparatus of claim 1, wherein the PLM has a first region and a second region, wherein the first region of the PLM is configured to receive the first light and the second region of the SLM is configured to receive the second light.
3. The apparatus of claim 2, wherein the first region of the PLM and the second region of the SLM are rectangular.
4. The apparatus of claim 2, wherein the first region of the PLM and the second region of the PLM are elliptical.
5. The apparatus of claim 1, wherein the SLM further has a third region, and the PLM is further configured to:
 - receive third light having a third color, wherein the third color is different than the first color and the second color; and
 - direct the third light towards the third region of the SLM.
6. The apparatus of claim 1, wherein the PLM is configured to scroll the first region of the SLM and the second region of the SLM using quasi-continuous steps.
7. The apparatus of claim 1, wherein the PLM is configured to scroll the first region of the SLM and the second region of the SLM using discrete steps.
8. The apparatus of claim 1, wherein the first region overlaps the second region.
9. The apparatus of claim 1, wherein the SLM is configured to produce an image in response to the first light and the second light.
10. The apparatus of claim 9, wherein the first region of the SLM is configured to produce a bright region of the image.
11. The apparatus of claim 1, further comprising:
 - a first light source configured to produce the first light; and

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a second light source configured to produce the second light.

12. An apparatus comprising:

- a first spatial light modulator (SLM);
- a second SLM; and
- a phase light modulator (PLM) optically coupled to the first SLM and to the second SLM, the PLM configured to:
 - receive first light having a first color;
 - receive second light having a second color, the second color different than the first color;
 - direct the first light towards the first SLM; and
 - direct the second light towards the second SLM.

13. The apparatus of claim 12, wherein the PLM has a first region and a second region, wherein the first region of the PLM is configured to receive the first light and the second region of the SLM is configured to receive the second light.

14. The apparatus of claim 13, wherein the first region overlaps the second region.

15. The apparatus of claim 12, wherein the first SLM is configured to produce a first image responsive to the first light and the second SLM is configured to produce a second image responsive to the second light.

16. A method comprising:

- receiving, by a first region of a phase light modulator (PLM), first light having a first color;
- receiving, by a second region of the PLM, second light having a second color, the second color different from the first color; and
- producing, by the PLM, a holographic image responsive to receiving the first light and the second color, wherein the holographic image has a first region and a second region, the first region of the holographic image having the first color and the second region of the holographic image having the second color.

17. The method of claim 16, wherein the PLM has a first region and a second region, wherein receiving the first light is performed in the first region and receiving the second light is performed in the second region.

18. The method of claim 16, wherein the first region of the PLM and the second region of the PLM are rectangular.

19. The method of claim 16, further comprising receiving, by the PLM, third light having a third color, wherein the third color is different than the first color and the second color, wherein producing the holographic image is further performed responsive to receiving the third light, and wherein the holographic image further has a third region having the third color.

20. The method of claim 16, wherein the first region of the holographic image overlaps the second region of the holographic image.

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